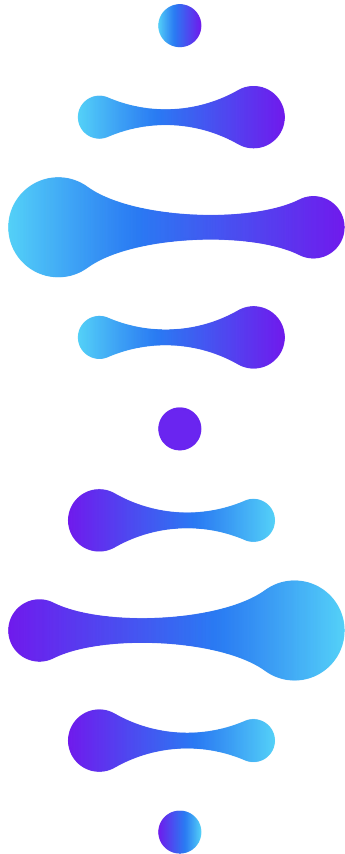




IntellxxDNA™

Intelligent Science. Intelligent Living.

Patient Report for:

Jenn Pei

DOB: 02/15/1994

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Prepared by:

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DISCLAIMER: IntellxxDNA™ (IXXD) is a Clinical Decision Support (CDS) tool to be used solely by licensed health care providers. The function of the tool is to present your patient's genomic information, in the form of a report, alongside references to peer-reviewed clinical studies relating to specifically chosen genetic variants. It will also provide referenced information about potential strategies that may help you address these panels.

IntellxxDNA™ is NOT meant to be used to diagnose or treat illnesses. The reports are intended to be one factor of clinical decision-making and to be used along with a full history, physical exam and laboratory tests as appropriate. Providers must review patient information and make their own independent clinical decisions regarding treatment.

Understanding your report

Your IntellxxDNA™ Report provides you and your clinician with the knowledge to help you achieve optimal health and wellness. This overview will help you understand the findings within your report.

Genomics vs. Genetics:

Genetics is the study of individual genes and how they are inherited. Genetic diseases arise if you inherit 1 copy of a gene from each parent, like cystic fibrosis and sickle cell anemia. Most chronic diseases (such as diabetes, heart disease, cognitive decline, and even autism and ADHD) are not conveyed by any one gene but are genomic in origin. Genomics is the study of small changes in DNA that by themselves are not disease-causing but can contribute to disease as they interact with each other along with diet, lifestyle, and the environment.

Alleles are one of two or more versions of a specific genetic sequence on a chromosome. Individuals inherit one allele from each parent for a particular gene. The minor allele is the less common version but may convey either benefit or risk.

Genes are like a set of instructions that tells your body how to make specific things, like proteins. These proteins play important roles in how your body grows, functions, and behaves.

Why this SNP is important What makes it significant and how it can affect you.

Gene Name	COMT V158M
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact (Odds Ratio)	Medium 1.73

"Catechol-O-Methyltransferase" gene

Your body does not break down the fight or flight molecules of adrenaline and norepinephrine as well. Too much of these stress molecules can increase the risk of a heart event.

- This contributes to a decreased ability to break down adrenaline and dopamine. Variants have also been associated with increased risk of stress, but also improved memory.
- This variant has been associated with an increased risk for acute coronary syndrome, which is having a heart event while running or under stress.
- If you have elevated homocysteine and this variant, it can further contribute to risk.

Supplements: Ashwagandha, rhodiola rosea, magnesium, zinc, EGCG, and genistein are supplements that may be beneficial and can be used with guidance on dosing, risks, and potential benefits.

Note: If you have two copies of this variant, you may want to avoid quercetin, as it can further decrease the breakdown of these fight or flight chemicals.

Prevalence is how common this allelic combination is in the population.

Impact is the likelihood and degree to which the variant may contribute to genomic disease. Some SNPs have an Odds Ratio, while others do not.

What you can do about it! Your clinician will help you decide what is right for you.

How this SNP works in your body. Some of the science behind the SNP.

Variant: AKA "SNP" is a slight difference or alteration in a gene that can affect how a gene functions and can potentially impact an individual's traits or health. A SNP is a one "letter" or nucleotide change in our DNA. SNPs occur naturally in a population or can arise from mutations. Different variants have different functions, which can make a gene more or less active or affect the quality of the proteins made.

ADHD Stimulant and Medication Response

The most common treatment for ADHD is stimulants (medications like Ritalin, Adderall, Vyvanse, etc.). Stimulants typically work by increasing levels of certain brain chemicals like norepinephrine and dopamine which are thought to play a role in attention and focus. It is important to consider that some people respond well to stimulants, while others do not. Almost one-third of people who take stimulants develop side effects from them or do not respond to treatment. Genomics is valuable to help determine if you specifically would respond to stimulants or not. This will also allow your clinician to develop a better treatment plan for you, as well as help decrease your risk of side effects if you do take a stimulant medication. If you do not respond well to stimulants, there are other lifestyle, supplement, and medication choices that can be utilized to help your ADHD symptoms. This panel is also beneficial for clinicians to consider treatment for children with Autism Spectrum Disorders (ASDs). In studies, the stimulant Methylphenidate (Ritalin) has been shown to reduce hyperactive-impulsive symptoms common in children with Autism Spectrum Disorders (ASDs), so using genomics can help identify if a child with Autism would benefit from it.

ADHD Stimulant and Medication Response Risk SNPs

Gene Name	SLC6A2
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	8.7% of people have this result
Impact	High

"Solute Carrier Family 6 Member 2" gene

You may respond less well to stimulants because you have a higher baseline of the chemical called norepinephrine (concentration brain chemical) which is what stimulants increase.

- This variant is associated with a decreased response to stimulants. You and your clinician can look at other panels like "Attention and Focus" and "Homocysteine" for specific recommendations for you if you have ADHD.

Gene Name	DBH T15791C
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	21.2% of people have this result
Impact	High

"Dopamine Beta-Hydroxylase" gene

You likely do not produce norepinephrine (concentration brain chemical) as well because you have less activity of an enzyme that converts dopamine (reward brain chemical) to norepinephrine. This can make you less likely to respond to medication like atomoxetine/Strattera.

- You are less likely to respond to the ADHD medication atomoxetine.

Lifestyle: Exercise may be beneficial at helping raise norepinephrine levels.

Supplements: Vitamin C supplementation may be beneficial as it is a cofactor for this enzyme.

Medications: Stimulants are an option for treating ADHD with this variant.

Gene Name	DRD2
Minor Allele	A
My Results	CA
My Variants	You have 1 minor allele
Prevalence	25.7% of people have this result
Impact	Medium

"Dopamine Receptor D2" gene

You may have higher levels of dopamine because your body does not have the ability to stop dopamine (reward brain chemical) from getting too high. This alteration appears to increase the risk of side effects with the stimulant, methylphenidate treatment, especially with higher doses.

- This variant is associated with an increased risk of substance use, depression, and higher teacher ADHD symptom ratings.
- This variant correlates with higher dosing methylphenidate being needed, as well as a higher risk of side effects to this medication.

You may want to avoid high-dose stimulants due to the potential for side effects.

Supplements: Bacopa monnieri can help lower dopamine levels. L-theanine, valerian, and lemon balm help improve the brain chemical GABA's activity. Increased GABA activity can help decrease expression of dopamine. These supplements can be considered with guidance on dosing, risks, and potential benefits.

Gene Name	SLC6A3 C17719T
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	36% of people have this result
Impact	Medium

"Solute Carrier Family 6 Member 2" gene

This gene is involved in dopamine (reward chemical) regulation by helping with dopamine transportation. You may get less improvement in ADHD symptoms with stimulants.

- You may notice less improvement of ADHD symptoms if you take a stimulant.

Supplements: Bacopa monnieri may be beneficial as there is evidence that it helps control dopamine response and it also has studies on improving ADHD symptoms. It can be considered with guidance on proper dosing.

Gene Name	SLC6A2 G47034T
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	45.5% of people have this result
Impact	Medium

"Solute Carrier Family 6 Member 2" gene

If you have 2 copies of this gene, you have increased signaling of the brain chemical norepinephrine (concentration). This makes you less likely to respond to medications like atomoxetine/Strattera that increase norepinephrine even more.

- Two variants are associated with a decreased response to ADHD medication atomoxetine (Strattera).
- If you have one variant, this is associated with ADHD, increased symptoms of hyperactivity, and challenges with your visual memory.

Recommendations with 2 variants: You and your clinician can look at other panels like "Attention and Focus" and "Homocysteine" for specific recommendations for you. **Recommendations with 1 (or 0) variant:** Atomoxetine, tricyclic antidepressants, bupropion, and stimulants are some ADHD medications that you may benefit from.

Gene Name	ABCB1
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	49.1% of people have this result
Impact	Medium

"ATP-binding cassette sub-family B member 1" gene

This variant makes the body remove certain medications more quickly, making them less effective. This can lead to drug resistance, especially with medications like guanfacine, morphine, and nortriptyline.

- AA (2 variants) is linked to a reduced response to guanfacine, making the medication less effective.

Medication: Your clinician may recommend avoiding guanfacine, as individuals with AA are less likely to respond to it for ADHD.

Gene Name	COMT V158M
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact	Medium

"Catechol-O-Methyltransferase" gene

You may have higher natural levels of dopamine (reward brain chemical), norepinephrine (concentration brain chemical), and adrenaline (fight or flight brain chemical). Because stimulants work by raising these same brain chemicals, and they are already high, you may be less likely to respond to stimulants.

- You may have a lower response to stimulants (methylphenidate).
- This increases your susceptibility to stress but also improves your ability to retrieve memories.
- If you have two copies of this variant you are less likely to respond to bupropion (medication prescribed for depression and ADHD medication).

Supplements: Magnesium, zinc, ashwagandha, and bacopa monnieri may help decrease the activity of the brain chemicals that you may have higher levels of and can be considered with guidance for proper dosing. You may want to avoid quercetin if you have two copies of this variant, as it may increase these brain chemical levels further.

ADHD Stimulant and Medication Response Benefit SNPs

Gene Name	DRD4 C8887A
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	25.4% of people have this result
Impact	Helpful

"Dopamine Receptor D4" gene

If you have 2 copies of this variant you are more likely to respond to stimulants because you may have a lower baseline level of dopamine (reward/motivation brain chemical) activity which stimulants work on.

- Having 2 variants is associated with an **increased** likelihood of responding to stimulants.
- If you have one variant you are **less likely** to respond to stimulants.

Gene Name	ADRA2A
Minor Allele	G
My Results	GC
My Variants	You have 1 minor allele
Prevalence	40.4% of people have this result
Impact	Helpful

"Adrenoceptor Alpha 2A" gene

You are more likely to respond to stimulants as stimulants activate this receptor that this variant acts on. This receptor is involved with coding for the chemicals adrenaline (fight or flight) and norepinephrine (concentration).

- Having 1 or 2 G variants has been associated with an **increased likelihood of responding to stimulants (methylphenidate)**. You may respond even more if you have 2 variants.
- This variant has been associated with social withdrawal behavior (avoiding people and social interactions).

Gene Name	SLC6A2
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	45.7% of people have this result
Impact	Helpful

"Solute Carrier Family 6 Member 2" gene
If you have 2 copies of this gene you may have more energy if you receive a stimulant.

- Two variants may lead you to feel more "elated" and full of energy with a high dose of a stimulant compared to a low dose.

Attention and Focus

Attention Deficit Hyperactivity Disorder (ADHD) is the most common neurodevelopmental disorder, affecting approximately 8-12% of children worldwide. An ADHD diagnosis is made by analyzing factors relating to developmentally inappropriate levels of inattention, hyperactivity and impulsivity. These ADHD associated behaviors have been shown to negatively affect performance in school and work. Interpersonal relationships may also suffer due to these symptoms.

There are a number of factors that contribute to ADHD, but genomics is thought to account for around 80% of cases. Social factors, as well as lifestyle factors such as nutrition, may also contribute. When it comes to genomics, things may seem complicated; many genes contribute to this disorder. However, this report focuses on the ADHD-associated variants with the largest impact. They help us understand the root causes of ADHD. Once these are investigated, we can gain a better understanding of how to treat this complicated disorder on a case-by-case basis. Some examples of an underlying cause of ADHD include difficulties in converting dopamine ("reward" brain chemical) to norepinephrine ("concentration and focus" brain chemical), low serotonin levels ("mood" brain chemical), and altered connectivity in the brain.

Genomics is NOT diagnostic. Having variants does NOT mean you have ADHD. Your licensed health care provider will discuss these variants with you and help you decide if you would benefit from any potential interventions. The results obtained in this report will assist your health care provider in developing a treatment plan specific to you.

Attention and Focus Risk SNPs

Gene Name	GCH1
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	4.1% of people have this result
Impact	High

"Guanosine Triphosphate Cyclohydrolase 1" gene

You may be less able to make a molecule called BH4 . BH4 helps make 3 important brain chemicals involved in attention and focus: norepinephrine (concentration and focus brain chemical), dopamine (motivation, reward, and movement brain chemical), and serotonin (mood brain chemical). Because of this you may make less of these brain chemicals. Two copies of this variant have been associated with decreased attention.

- 2 variants may lead to you having decreased attention ability.
- 2 variants also are associated with lower nitric oxide levels. Nitric oxide helps keep your blood vessels open.

Diet: Improving nitric oxide synthesis may be beneficial. Leafy greens and beets are examples of foods that support nitric oxide synthesis. You can ask your clinician for the "Nitric Oxide producing foods" found in the document library.

Supplements: Methylfolate helps increase BH4 levels. Other supplements that may be beneficial include vitamin C, pycnogenol, curcumin, and ginkgo biloba and can be considered with guidance on dosing.

Note: BH4 supplementation has been studied as a benefit for Autism; however it can be expensive. If you have two copies of this comorbid with Autism it may be worth considering.

Gene Name	MAOA (R297R)
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	11.7% of people have this result
Impact	High

"Monoamine Oxidase A" gene

You may have increased amounts of monoamine oxidase A (MAO enzyme), which helps to break down dopamine (motivation, reward, and movement brain chemical), norepinephrine (concentration, and focus brain chemical), and serotonin (mood brain chemical). This can lead to decreased amounts of norepinephrine and dopamine. This can contribute to ADHD.

- You may have an increased risk of ADHD.
- **This variant is located on the X-chromosome, so it is important if you have two copies if you are female, and one copy if you are male.**

Diet: Some food that may be beneficial includes peppermint, tangerines, some types of coffee, and nutmeg.

Supplements: Gingko biloba may be beneficial as it is a natural MAO inhibitor, and can be considered with guidance on dosing. Vitamin C increases norepinephrine.

Medications: Stimulants and bupropion may help improve ADHD symptoms as they both increase norepinephrine and dopamine.

Gene Name	ADGRL3
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	15% of people have this result
Impact	High

"Adhesion G Protein-Coupled Receptor L3" gene

You may have less of a gene that is responsible for activating certain areas of the brain. This can lead to ADHD symptoms like hyperactivity and impulsivity. You may respond better to stimulants, however.

- This variant has been associated with low glutamate (excitatory brain chemical).
- This variant increases your risk of ADHD, more so if you have two copies. In addition, children with this variant may develop ADHD earlier than other people (before age 7).
- There have been more studies on ADHD risk with males.
- You may have a better response to stimulants when prescribed for ADHD.

Lifestyle: Biofeedback/neurofeedback may help improve ADHD symptoms.

Supplements: Citicoline is a supplement that you may benefit from and can be considered with guidance on dosing.

Medications: If you have ADHD, a stimulant may be more effective for you.

Gene Name	DBH
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	24% of people have this result
Impact	High

"Dopamine Beta-Hydroxylase" pathway.

You are more likely to have lower levels of norepinephrine (concentration and focus brain chemical) and higher dopamine (motivation, reward, and movement brain chemical). This can lead to an increased risk of ADHD.

- This variant may increase your risk of ADHD by as much as 15 times if you have this in the combination with the variant below (rs6271).
- Vitamin C is a cofactor (helper) for this enzyme.

Lifestyle: Exercise may be beneficial as it can raise norepinephrine levels.

Supplements: Ginkgo Biloba, vitamin C, copper, zinc, methylfolate, and certain probiotics (saccharomyces and bacillus) are supplements that you may benefit from and can be considered with guidance on dosing.

Medications: A stimulant may help improve ADHD symptoms.

Gene Name	ANKK1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	32.1% of people have this result
Impact	Medium

"Ankyrin Repeat and Kinase Domain Containing 1" gene

Your body may not decrease dopamine in response to elevated levels as well as someone without this gene variant. This can lead to higher levels of dopamine (motivation, reward, and movement brain chemical). This is associated with an increased risk of ADHD.

- This variant is associated with an increased risk of ADHD.
- In addition to ADHD, this variant may increase your risk of food addiction, and obsessive-compulsive disorder (OCD).

Lifestyle: You may benefit from moderate to intense exercise or from eating foods that are high in butyrate (asparagus, onions, garlic are examples). You can ask your clinician for the "butyrate-producing foods" handout found in the document library.

Supplements: Bacopa monnieri helps decrease dopamine levels, and citicoline may also be beneficial. These supplements can be considered with guidance on dosing.

Gene Name	SLC6A3
Minor Allele	C
My Results	GC
My Variants	You have 1 minor allele
Prevalence	35.9% of people have this result
Impact	Medium

"Solute Carrier Family 6 Member 3" gene

You are likely to have higher levels of dopamine (reward and movement brain chemical). This can increase the risk of ADHD.

- This variant increases your risk of ADHD.

Supplements: Bacopa monnieri helps decrease dopamine signaling, and can be considered with guidance on dosing.

Medication: Stimulants may be beneficial; however you may be more likely to experience "euphoria" after a stimulant.

Gene Name	HTR1B
Minor Allele	G
My Results	CG
My Variants	You have 1 minor allele
Prevalence	40% of people have this result
Impact	Medium

"5-Hydroxytryptamine Receptor 1B" gene

Your serotonin (mood brain chemical) receptors may be more responsive. This also increases glutamate (excitatory brain chemical), acetylcholine (memory brain chemical), and dopamine (motivation, reward, and movement brain chemical), and lowers GABA (calm brain chemical). This variant may increase the risk of childhood ADHD.

- You may find that it is more difficult to stop a task once starting it.
- It is also important to be aware that this variant is also found to be associated with alcoholism, antisocial personality traits, and intermittent explosive disorder.

Supplements: NAC, magnesium, citicoline, and vitamin B6 can help lower glutamate. Ashwagandha, L-theanine, vitamin B6, passion flower, valerian, and lemon balm may help raise GABA levels. These supplements can be considered with guidance on dosing, risks, and potential benefits.

Note: SSRIs and 5-HTP may need to be avoided as you are less likely to respond to them and they actually could have the potential of making your symptoms worse.

Gene Name	CNTNAP2
Minor Allele	T
My Results	AT
My Variants	You have 1 minor allele
Prevalence	46.6% of people have this result
Impact	Medium

"Contactin Associated Protein Like 2" gene

You may have difficulty with auditory and verbal learning and processing due to changes in your brain connections. This also can correlate to less GABA (calm brain chemical) release, which can contribute to hyperactivity symptoms.

- This variant increases your risk of ADHD.
- Two variants have been associated with Autism.

Lifestyle: You may benefit from learning styles that are not auditory (experiential, visual, etc.). Biofeedback and or neurofeedback have evidence in improving ADHD symptoms.

Supplements: Citicoline, L-theanine, and Vitamin D may be beneficial to help decrease ADHD symptoms and/or improve GABA levels and may be considered with guidance on dosing.

Gene Name	CNTFR
Minor Allele	T
My Results	TG
My Variants	You have 1 minor allele
Prevalence	48.5% of people have this result
Impact	Medium

"Ciliary Neurotrophic Factor Response" gene

This gene relates to the production and protection of brain cells (neurons), which helps with attention. You may have changes in this pathway which can lead to inattentive ADHD symptoms.

- This variant may be linked to more severe inattentive symptoms with ADHD.
- Supplements:** Inositol, citicoline, NAC, magnesium, and vitamin B6 are examples of supplements that may be beneficial and can be considered with guidance on dosing.

Gene Name	COMT V158M
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact	Medium

"Catechol-O-Methyltransferase" gene

Your body likely takes longer to break down adrenaline (energy brain chemical), norepinephrine (concentration and focus brain chemical), and dopamine (motivation, reward, and movement brain chemical). If you have two copies of this you may have even higher levels of these brain chemicals which can contribute to ADHD.

- If you are male you may have an increased risk of ADHD and have a decreased response to stimulants.
- Two variants has showed to relate to difficulties with working in a cooperative setting.

Supplements: If you have two variants ashwagandha may be beneficial. Magnesium can boost this enzyme's (COMT) ability. Zinc can also be considered. These supplements can be considered with guidance on dosing. If you have two variants, you may want to avoid quercetin as it can slow down this enzyme's activity even further.

Gene Name	FADS2
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	46.8% of people have this result
Impact	Medium

"Fatty Acid Desaturase 2" gene

EPA and DHA are crucial omega 3 fatty acids needed for attention and cognitive focus. If you have 2 copies of this gene, your body likely has more natural EPA/ DHA production. This can decrease ADHD symptoms.

- Two variants are associated with a decreased risk of ADHD.
- One variant associated with increased ADHD risk and lower omega-3 levels.

Lifestyle for 1 (or 0) variant: Avoid alcohol as it can further lower omega-3 levels.
Supplements for 1 (or 0) variant: EPA and DHA omega-3 fish oils may be beneficial and can be considered with guidance on dosing.

Attention and Focus Benefit SNPs

Gene Name	SLC6A2 G47034T
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	45.5% of people have this result
Impact	Helpful

"Solute Carrier Family 6 Member 2" gene

Norepinephrine is a brain chemical responsible for concentration and focus. If you have 2 copies of this gene, your body likely has more norepinephrine available. This can decrease ADHD symptoms.

- Two variants are associated with a decreased risk of ADHD.
- If you have only one variant (or 0) you may benefit from atomoxetine, a medication prescribed for ADHD.

Lifestyle for 1 (or 0) variant: Exercise can help support raising norepinephrine levels.

Supplements for 1 (or 0) variant: Ginkgo biloba and vitamin C may be beneficial and can be considered with guidance on dosing.
Medications for 1 (or 0) variant: Atomoxetine, Bupropion, and potentially stimulants are medications that can be considered for ADHD.

Gene Name	DRD4
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	50.8% of people have this result
Impact	Helpful

"Dopamine Receptor D4" gene
If you have two copies of this gene variant you may have a decreased ADHD risk due to you having a better functioning dopamine receptor (dopamine is responsible for reward and motivation).

- If you have two variants you are less likely to have ADHD than someone without this variant.

Celiac

Celiac disease is an extreme form of gluten sensitivity, characterized generally by abdominal discomfort, inflammation, malabsorption, and gastrointestinal symptoms as well as other effects throughout the body. Full celiac disease can be diagnosed by biopsies done by endoscopy (utilizing a scope that goes into the stomach and small intestines) and through blood work.

People without celiac disease may still have gluten sensitivity. In these cases, people release more of a compound called "zonulin" which can lead to increased gut permeability or a leaky gut. This can then contribute to brain fog, muscle and joint pain, fibromyalgia, and nerve symptoms as well as rashes.

Celiac disease can have a strong autoimmune component, but interestingly people are more likely to develop autoimmune diseases if they develop Celiac disease later in life.

Celiac genomics can be confusing and complex. The top SNP listed in the report (will be shown in red if present), the SNP for HLA 2.5, is the main contributor for celiac disease risk and is found in 90% of true celiacs. However, having this gene DOES NOT mean you will get celiac disease. Only about 35% of individuals with this variant have/will get full celiac. Others however do have a significantly higher risk of being gluten intolerant.

Other genes that influence the risk of celiac disease often work in combination to increase risk. This is called a haplotype. In order to be at higher risk from the HLA 2.2 haplotype you have to not only have the HLA-DQ2.2 risk SNP, but also not have the 2 benefit variants for the HLA-DQ4 and HLA-DQ2.2.

Other genes that affect gut permeability and autoimmunity also contribute to gluten sensitivity and celiac risk.

Celiac Risk SNPs

Gene Name	CTLA4
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	13.3% of people have this result
Impact (Odds Ratio)	 Medium 1.24

"Cytotoxic T-Lymphocyte Associated Protein 4" gene

This protein helps put "brakes" on our system for the body's immune response, which helps stop an immune response. You may be less able to turn off your immune response which can lead to an increased risk of celiac disease, as well as other autoimmune conditions.

- This variant is associated with an increased risk of celiac disease. It also may lead to an increased risk of Grave's disease, type 1 diabetes, and alopecia areata (patchy hair loss).

Diet: Speak with your clinician in advance about a trial of a gluten free diet, especially if having symptoms. You can ask your clinician for the Go Gluten-Free for Gluten Sensitive or Go Gluten Free for True Celiac handouts that can be found in the Document Library if this applies to you.

Supplements: Curcumin and lignite extract may be beneficial with guidance on dosing.

Gene Name	HLA-DQ2.2
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	24.1% of people have this result
Impact (Odds Ratio)	High 2.56

"Human Leukocyte Antigen (DQ2.2)" gene

You are more likely to present gluten to your immune system as if it were a foreign invader. This may increase your risk of developing gluten intolerance or celiac disease if you do not have HLA benefits listed below in green (or if you have HLA-DQ2.5 above).

- This variant is significant if in a combination of HLA-DQ2.5 or with a lack of benefits in HLA-DQ2.2 or HLA-DQ4 below.

Diet: If full combination of 3 SNPs and if symptomatic, speak with your clinician in advance about a trial of a gluten-free diet. It may help alleviate symptoms. Can have gluten sensitivity with a negative celiac test. You can ask your clinician for the Go Gluten-Free for Gluten Sensitive or Go Gluten Free for True Celiac handouts that can be found in the Document Library, if this applies to you.

Note: If you have any symptoms of celiac disease it may be beneficial to be tested for celiac.

Gene Name	LPP
Minor Allele	A
My Results	CA
My Variants	You have 1 minor allele
Prevalence	48.9% of people have this result
Impact (Odds Ratio)	Medium 1.47

"LIM Domain Containing Preferred Translocation Partner in Lipoma" gene

This variant can cause you to have changes to shape in a protein involved in protecting your gut barrier. This may increase your risk for gluten sensitivity and celiac disease.

- This variant is associated with increased celiac disease risk.

Supplements: L-glutamine and lignite are supplements that can help protect your blood gut barrier and can be considered with guidance on dosing.

Gene Name	MYO9B
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	49.5% of people have this result
Impact (Odds Ratio)	Medium 1.36

"Myosin IXB" gene

You may make more of something called "zonulin" which contributes to a "leaky gut". Higher zonulin levels can also contribute to celiac disease and inflammatory bowel issues.

- Risk of celiac disease is increased more based on studies if you have two variants.
- This variant has also been associated with an increased risk of inflammatory bowel disease.

Diet: Speak with your clinician in advance about a trial of a gluten free diet, especially if having symptoms. You can ask your clinician for the Go Gluten-Free for Gluten Sensitive or Go Gluten Free for True Celiac handouts that can be found in the Document Library if this applies to you.

Supplements: Various supplements like lignite extract, serum-derived bovine immunoglobulins, berberine, l-glutamine, and colostrum may be beneficial for this pathway and can be considered with guidance on dosing.

Choline

Choline, which is a precursor of phosphatidylcholine, is an essential nutrient for many organs including the brain. Your nerves, nerve sheaths, and brain require enough phosphatidylcholine in order to properly carry electrical signals. Low levels have been associated with poor memory and Alzheimer's. Choline is also needed for proper liver function, and decreased amounts can contribute to elevated liver functions and fatty liver. Citicoline (CDP-Choline) supplementation has been shown to be beneficial for a wide range of neurological issues including cognition, post-stroke, repairing of damaged nerves, and even strabismus (cross-eyed). Because there is a positive interaction between estrogen and choline synthesis pathways, postmenopausal women, infants, and children, and some men are at higher risk of choline deficiency. Adequate choline intake is also important for pregnant and lactating women as it is critical during fetal development of the brain and spinal cord.

This panel looks at slight changes in DNA called SNPs or variants that may impact your choline levels or increase your risk of certain things like muscle damage when your choline levels are low.

Choline Risk SNPs

Gene Name	PEMT
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	48.5% of people have this result
Impact	Medium

"Phosphatidylethanolamine N-Methyltransferase" gene

This variant may lead to decreased activity of an enzyme that is necessary to synthesize choline. This can decrease your levels of choline and therefore, phosphatidylcholine. Phosphatidylcholine is essential for nerve and brain repair.

- Estrogen increases choline production. Due to this, men and post-menopausal women are at higher risk with this variant.
- If your choline levels are not optimized, you may have an increased risk of cognitive impairment, and fatty liver disease.

Diet: Increasing dietary choline intake may help. Examples include beef liver, soybeans, egg etc. Other foods are found in the "Choline Containing Foods" handout that you can ask your clinician for.

Supplements containing choline or phosphatidylcholine also may be beneficial and can be considered with guidance on dosing.

Gene Name	MTHFD1 G1958A
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	48.8% of people have this result
Impact	Medium

"Methylenetetrahydrofolate Dehydrogenase, Cyclohydrolase And Formyltetrahydrofolate Synthetase 1" gene

This enzyme is needed for the conversion of folic acid to its active form and is important for the building blocks for DNA synthesis and repair. This is associated with a 50% reduction in this enzyme's activity. This variant is also associated with lower levels of choline if dietary choline is not sufficient.

- If your folate levels are not maintained during pregnancy you may have an increased risk of neural tube defects.
- Women with variants in this pathway may need twice the usual amounts of choline to help repair their DNA. In studies, 930 mg choline a day were used instead of typical 480 mg/day.

Diet: Incorporating foods with Folate (spinach), Vitamin B12 (clams), Betaine (beets) and/or Choline (eggs) may be beneficial. For more suggestions, you can ask your clinician for handouts on these nutrients that are found in the document library.

Supplements: Folic acid, folinic acid, or other forms of folate may be beneficial. Choline, betaine, and B12 can also be of benefit and can be considered with guidance of dosing.

Choline Benefit SNPs

Gene Name	CHDH
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	1.7% of people have this result
Impact	Helpful

"Choline Dehydrogenase" pathway.

You have a decreased risk for choline deficiency because you have a better ability to "store" choline.

- This variant is also associated with a better ability to detoxify from arsenic.
- Despite adequate choline, you may need an increased amount of betaine. Betaine is a chemical in your body that helps with liver function and helps metabolize homocysteine.

Diet for 1 or 2 variants: Incorporating betaine (beets, shrimp, spinach, etc.) into your diet may be beneficial. You can ask your clinician for the "Betaine Containing Food" handout that can be found in the Document Library.

Supplements for 1 or 2 variants: Betaine may be beneficial.

Gene Name	BHMT R239Q
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	9.3% of people have this result
Impact	Helpful

"Betaine-Homocysteine S-methyltransferase" gene

If you have 2 copies of this variant, you are more likely to synthesize phosphatidylcholine over betaine from choline. Phosphatidylcholine is essential for nerve and brain repair. This also may make you have a better ability to recycle homocysteine to methionine.

- This variant may decrease your risk of heart disease, certain neural tube defects (spina bifida), and cervical cancer.
- If you are a post-menopausal woman with two copies of this variant, studies indicate you have a decreased risk of breast cancer.

Diet for 1 (or 0) variant: Increasing dietary choline intake may help. Examples include beef liver, soybeans, egg, etc. You can ask your clinician for the Choline Containing Foods handout that can be found in the Document Library. In addition, you may benefit from a diet that is higher in methionine (chicken, fish, etc.).

Supplements 1 (or 0) variant: Choline or phosphatidylcholine also may be beneficial and can be considered with guidance on dosing.

Copper and Zinc

Zinc is an essential nutrient and is needed in over 2000 reactions for your body. It is essential to help you fight infections and to maintain a healthy gut, brain, and all the nerves in your body. Zinc also helps decrease inflammation. ***Zinc is needed in the immune system, brain development, vision, wound healing, and the activity of over 300 different reactions in the body.***

Zinc deficiency is associated with decreased cognitive functions and behavioral symptoms such as irritability, fatigue, and depression. There are some studies suggest that zinc supplementation and restoration of normal/ideal zinc levels can protect against cognitive decline.

Copper is essential for the function of many enzymes (proteins that assist in chemical reactions in the body). In the brain, copper is needed for generating energy, immune defense, and iron metabolism. Copper deficiency is associated with neurological (brain and nerve) problems such as nerve damage to your extremities, or eye damage.

The copper to zinc ratio typically increases with age and has been found to be elevated among the elderly with Alzheimer's disease, stroke, Parkinson's disease, and other conditions. However, when it comes to copper and zinc, there is a happy medium. Too high of zinc levels and too low of copper (or vice versa) is also not good for the brain, nerve function, and can further contribute to Alzheimer's and poor cognition. In a study, the ideal copper to zinc ratio for brain function appears to be approximately 1:1. The panel looks at slight changes in DNA called SNPs or variants. Potential interventions can be targeted to the specific variants that you have.

Copper and Zinc Risk SNPs

Gene Name	NBDY
Minor Allele	C
My Results	CT
My Variants	You have 1 minor allele
Prevalence	43.2% of people have this result
Impact	Medium

"Negative Regulator of P-body Association" gene

Zinc is needed in the immune system, brain development, vision, wound healing, and the activity of over 300 different reactions in the body. Having 2 variants has been associated with lower zinc levels.

- Because this gene is on the X-chromosome, males that have 1 variant will be shown as having 2 variants.

***Diet for 2 variants:** Eating a high zinc diet may be beneficial. High zinc foods include oysters (the highest of all foods), liver, and other meats.*

***Supplements for 2 variants:** Zinc supplementation may be beneficial, particularly if brain health concerns or if two variants. Monitoring of zinc can be done with blood work.*

Copper and Zinc Benefit SNPs

Gene Name	SLC30A8
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	41.1% of people have this result
Impact	Helpful

"Solute Carrier Family 30 Member 8" gene
This is a zinc transporter in pancreatic beta cells, and this SNP correlates with better ability to keep zinc in the pancreas. Higher levels of pancreatic zinc can increase insulin (a hormone that helps lower blood sugar) levels.

- This variant has been associated with decreased risk of diabetes, especially if zinc levels were in the ideal range.

Gene Name	CA1
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	47.5% of people have this result
Impact	Helpful

"Carbonic Anhydrase 1" gene
Variants may increase activity of a zinc-containing enzyme that is involved in a variety of biological processes.

- This variant is associated with higher levels of zinc.

Histamine Intolerance and Food Sensitivity

Histamine is broken down by the two enzymes diamine oxidase (DAO) and histamine N-methyltransferase (HNMT). HNMT is important in the brain and lungs but does not correlate with food histamine intolerance. Diamine oxidase is the main enzyme that breaks down histamine and other products of protein degradation from food. Additionally, some medications, strains of bacteria and certain foods can increase histamine release from "mast cells" found in the gut lining. If histamine is not broken down, high levels can lead to symptoms including: diarrhea and GI upset, headaches and migraines, low blood pressure, muscle pain, hives, rapid heart-beat or sense of panic, and many other symptoms.

The risk SNPs associated with the DAO enzyme correlate to low enzyme activity. This predisposes individuals to increased risk of histamine intolerance, but gene variants alone are not sufficient to cause clinical histamine intolerance. Manifestation of symptoms often requires additional triggers. One common trigger is a damaged intestinal lining. During times of inflammation, such as after traveler's diarrhea or in the presence of small intestinal bacterial or yeast overgrowth or with other inflammation, symptoms may be worse.

There are supplements, specific probiotics and dietary changes that can help prevent and minimize symptoms. Fish and meats that are old or have been left out are known to have the highest amounts of histamine. Canned and fermented foods as well as champagne and many fruits and vegetables are also high in histamine.

The 2 MCM6 SNPs found in the benefit SNPs section correlate to whether or not an individual is more likely to develop lactose intolerance as an adult. The ABSENCE of beneficial variants (i.e. 0 variants or "no green") is associated with hypolactasia and predicts the decline of intestinal lactose, thus decreasing ability to digest lactose found in milk and other dairy foods. If experiencing GI distress, your clinician may recommend decreasing dietary dairy intake to determine whether or not lactose malabsorption is contributing to gut symptoms.

The two HLA SNPs in the risk SNPs section relate to a significant genetic risk for developing a peanut allergy.

Histamine intolerance is a well-recognized cause of GI distress in the UK and Australia, but much less recognized in the United States.

Histamine Intolerance and Food Sensitivity Risk SNPs

Gene Name	AOC1 (ABP1)
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	34.8% of people have this result
Impact	Medium

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have increased risk for "histamine intolerance", which often manifests as GI symptoms.

- T allele has been associated with 1.82 times the risk of histamine intolerance in individuals with low enzyme activity.
- You may have decreased ability to break down histamine in the gut due to decreased production/activity of the enzyme diamine oxidase.

Diet: You can ask your clinician for the Histamine Intolerance handout which provides a list of food and drinks that are high in histamine and can be found in the Document Library.

Supplements: Diamine oxidase enzyme supplementation taken with high histamine foods has been shown to decrease symptoms 50-70%. Other supplements that may help reduce histamine release include quercetin, bromelain, pycnogenol, luteolin, and a probiotic with *L. rhamnosus*. These can be considered with guidance on dosing.

Gene Name	AOC1 (ABP1)
Minor Allele	T
My Results	AT
My Variants	You have 1 minor allele
Prevalence	37.3% of people have this result
Impact	Medium

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have increased risk for "histamine intolerance", which often manifests as GI symptoms.

- T allele has been associated with reduced diamine oxidase (DAO) enzyme activity.
- You may have decreased ability to break down histamine in the gut due to decreased production/activity of the enzyme needed to degrade it.
- If 2 variants in this pathway, you may have approximately double the risk of histamine intolerance compared to people with 1 variant.

Diet: You can ask your clinician for the Histamine Intolerance handout which provides a list of food and drinks that are high in histamine and can be found in the Document Library.

Supplements: Diamine oxidase enzyme supplementation taken with high histamine foods has been shown to decrease symptoms 50-70%. Other supplements that may help reduce histamine release include quercetin, bromelain, pycnogenol, luteolin, and a probiotic with *L. rhamonosus*. These can be considered with guidance on dosing.

Gene Name	AOC1 (ABP1)
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	37.4% of people have this result
Impact	Medium

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have increased risk for "histamine intolerance", which often manifests as GI symptoms or migraines. This may also manifest as increased sensitivity to non-steroidal anti-inflammatory (NSAIDs) medications like ibuprofen and naproxen.

- T allele has been associated with reduced diamine oxidase (DAO) enzyme activity.
- You may have decreased ability to break down histamine in the gut due to decreased production/activity of the enzyme needed to degrade it.
- T allele has been associated with 1.61 times the risk of migraines and even higher rates of migraines seen in women.

Diet: You can ask your clinician for the Histamine Intolerance handout which provides a list of food and drinks that are high in histamine and can be found in the Document Library.

Supplements: Diamine oxidase enzyme supplementation taken with high histamine foods has been shown to decrease symptoms 50-70%. Other supplements that may help reduce histamine release include quercetin, bromelain, pycnogenol, luteolin, and a probiotic with *L. rhamonosus*. These can be considered with guidance on dosing.

Gene Name	HLA-DQA2
Minor Allele	C
My Results	CT
My Variants	You have 1 minor allele
Prevalence	43% of people have this result
Impact	Medium

"Major Histocompatibility Complex, Class II, DQ Alpha 2" gene

Individuals with this SNP are more likely to develop peanut allergies due to how they present these allergens to the immune system. This allergy can be acquired during development.

- C allele has been significantly associated with 1.7 times the risk of developing a peanut allergy in Caucasians.
- This SNP has been associated as being the most significant for developing a peanut allergy.

Diet: You can speak with your clinician for suggestions to minimize the risk for developing a peanut allergy. **Important note: To be done under direct clinical supervision.**

Gene Name	HLA-DRA
Minor Allele	T
My Results	TG
My Variants	You have 1 minor allele
Prevalence	45.3% of people have this result
Impact	Medium

"Major Histocompatibility Complex, Class II, DR Alpha" gene

Individuals with this SNP are more likely to develop peanut allergies due to how they present these allergens to the immune system. This allergy can be acquired during development.

- T allele has been associated with 1.8 times the risk of a peanut allergy in Caucasians.
- You may have an increased risk of developing a peanut allergy.

Diet: You can speak with your clinician for suggestions to minimize the risk for developing a peanut allergy. **Important note: To be done under direct clinical supervision.**

Histamine Intolerance and Food Sensitivity Benefit SNPs

Gene Name	AOC1 (ABP1)
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	18.6% of people have this result
Impact	Helpful

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have higher enzyme levels and have decreased risk for "histamine intolerance".

- C allele has been associated with increased diamine oxidase (DAO) enzyme activity.
- You may have protective effects/increased ability to break down histamine in the gut due to increased production/activity of the enzyme necessary to degrade it.

Gene Name	AOC1 (ABP1)
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	32.6% of people have this result
Impact	Helpful

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have higher enzyme levels and have decreased risk for "histamine intolerance".

- "A" allele has been associated with increased diamine oxidase (DAO) enzyme activity.
- You may have protective effects/increased ability to break down histamine in the gut due to increased production/activity of the enzyme necessary to degrade it.

Gene Name	AOC1 (ABP1)
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	33% of people have this result
Impact	Helpful

"Amine Oxidase Copper Containing 1" gene

This enzyme is needed for breaking down histamine from foods. Individuals with this SNP have higher enzyme levels and have decreased risk for "histamine intolerance".

- "A" allele has been associated with increased diamine oxidase (DAO) enzyme activity.
- You may have protective effects/increased ability to break down histamine in the gut due to increased production/activity of the enzyme necessary to degrade it.

Homocysteine and Methylation

Homocysteine is made from the dietary amino acid methionine. Methionine is found in all sources of protein, with the highest levels noted to be found in fish and chicken, followed by red meat, dairy, and eggs. The lowest levels are found in plant sources of protein like nuts and seeds. Elevated levels of homocysteine have been associated with a wide variety of cardiac and neurological conditions including heart disease, stroke, Alzheimer's/impaired cognition, ADHD, migraines and spina bifida risk. Elevated levels of homocysteine can irritate the lining of the heart and is known to have detrimental effects on the brain. This increases the risk of cardiac events and strokes that are associated with dementia, as well as neurological events such as headaches and migraines.

Elevated homocysteine levels can be addressed in a targeted fashion by utilizing genomics. There are three main ways to clear homocysteine: the folate/methylfolate and vitamin B12 dependent pathway (MTHFR, MTR (also called MTS), MTRR), the choline dependent pathway (BHMT), and the vitamin B6 dependent pathway (CBS). Methylfolate (active folate) is also of great importance for the brain to make a variety of neurotransmitters. This panel also maps out the famous MTHFR (methylenetetrahydrofolate reductase) SNPs involved in making the neurotransmitters that are key for memory and mood/depression. There are a wide variety of B vitamins including folate/methylfolate, vitamin B6, B12, as well as supplements such as betaine, choline and cysteine that can be used to target genomics and reduce homocysteine, and potentially decrease the risk of homocysteine related conditions.

Homocysteine and Methylation Risk SNPs

Gene Name	MTHFR A1298C
Minor Allele	G
My Results	TG
My Variants	You have 1 minor allele
Prevalence	42.4% of people have this result
Impact	Medium

"Methylenetetrahydrofolate Reductase" gene

MTHFR is one of the more famous SNPs. It is needed to make the brain chemicals of serotonin (mood), norepinephrine (attention), dopamine (pleasure), melatonin (sleep) and acetylcholine (memory). MTHFR also plays an important role in DNA methylation, which is the way our body turns genes off.

- Variant (G) is associated with decreased MTHFR activity and elevated homocysteine.
- This variant increases the risk of ischemic stroke, male infertility, and depression.

Diet: Choline can also be a benefit, so ask your clinician about the Choline Containing Foods handout that can be found in the Document Library.

Supplements: Methylfolate can be of benefit for this SNP to help your ability to make these brain chemicals. Optimal levels of Vitamin B12 and Vitamin C are necessary for the methylfolate to function its best.

Gene Name	MTHFD1 G1958A
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	48.8% of people have this result
Impact	Medium

"Methylenetetrahydrofolate Dehydrogenase, Cyclohydrolase And Formyltetrahydrofolate Synthetase 1" gene

This SNP has three functions including helping make the building blocks for DNA synthesis and DNA repair. This SNP is also needed for the conversion of folic acid to its active form (methylfolate) for use by the MTHFR enzyme.

- Enzyme important for the making of the amino acid derived DNA building blocks that are needed for DNA synthesis and repair in addition to protein synthesis.
- Variant is associated with
 - Elevated homocysteine.
 - Increased susceptibility to choline deficiency.
 - Immune deficiencies
 - Neural tube defects

Supplements: Folinic acid or folate may be preferred replacement form for this variant along with B12 rather than just methylfolate in addressing this pathway as it is an earlier step before "MTHFR"

Diet: See the Betaine Containing Foods handout that can be found in the Document Library, as betaine may be beneficial for this pathway.

Gene Name	SLC19A1 (RFC)
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	49.3% of people have this result
Impact	Medium

"Reduce Folate Carrier" gene

This SNP reduces the body's ability to absorb folate, leading to lower folate and thiamine levels. It has been linked to a higher risk of congenital heart disease and increased homocysteine levels.

- Variants associated with congenital heart disease and can contribute to elevated homocysteine by decreasing cellular folate access.

Supplements: Methylfolate and thiamine are examples of interventions for this pathway as they allow folate to bypass reduced transport and help maintain proper cellular function, while thiamine supports energy metabolism and nerve function, reducing risks linked to this SNP.

Homocysteine and Methylation Benefit SNPs

Gene Name	BHMT R239Q
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	9.3% of people have this result
Impact	Helpful

"Betaine-Homocysteine S-Methyltransferase" gene

Individuals with two copies of this SNP are associated with having lower levels of homocysteine, and generally do not get significant homocysteine reduction with folate. Individuals with no variants may need a higher methionine diet and may have higher homocysteine levels.

- BHMT is a zinc-dependent enzyme that uses betaine to clear homocysteine.
- Individuals with this variant are associated with lower homocysteine levels.

Diet: If 1 variant, discuss with your clinician optimal amounts of betaine and choline. See the Betaine and Choline Containing Foods handout that can be found in the Document Library.

Gene Name	DPEP1
Minor Allele	C
My Results	GC
My Variants	You have 1 minor allele
Prevalence	34.5% of people have this result
Impact	Helpful

"Dipeptidase 1" gene.
Individuals with this SNP over-excrete cysteine resulting in greater conversion of homocysteine to cysteine, lowering homocysteine levels.

- You may have protective effects with lower levels of homocysteine.

Inflammation

Inflammation and immune system dysregulation are hallmarks of neurological and other various disorders. For example, biomarker analysis has demonstrated that various pro-inflammatory molecules are increased in autistic children (brain, CSF, and periphery), while anti-inflammatory molecules are decreased.

Tumor necrosis factor alpha (TNF α) is a pro-inflammatory cytokine capable of crossing the blood brain barrier, and has consistently been shown to be elevated in individuals with Alzheimer's. TNF α is expressed in various cell types including microglia in the brain, neurons, and astrocytes, and is capable of exerting a wide range of physiological changes (induce expression of other proinflammatory cytokines via NF κ B and other signaling pathways). Chronic neuroinflammation is often propagated by constitutively activated microglia, astrocytes, and neurons. Aberrant signaling can then lead to excitotoxicity, neuronal death and synaptic loss.

Interleukin 1 beta (IL1B), interleukin 1 alpha (IL1A), interleukin 6 (IL6), and interleukin 17 (IL17) are also pro-inflammatory cytokines shown to be elevated in a variety of disease states. IL17, for example, has been shown in studies to be elevated in psoriasis skin lesions. Interleukin 10 (IL10) on the other hand is an anti-inflammatory cytokine that is typically decreased in Autism Spectrum Disorder (ASD). These signaling molecules are highly intertwined and can regulate each other's production. If proper safeguards are not maintained and inflammation is not resolved, the inflammatory signaling cascade can spiral out of control and can contribute to a cytokine storm.

C-reactive protein (CRP) is an "acute phase reactant" that rises rapidly in response to tissue insult or injury, and can increase by as much as 1000-fold from baseline within 24 hrs. CRP is synthesized primarily by the liver upon interleukin 6 (IL6) stimulation. CRP is involved in inflammation and mediates activation of the complement system as well as cell mediated immunity via its interaction with the "Fc" receptors of Immunoglobulin G. On a genomic level, CRP regulation/production can be altered via polymorphisms that affect IL6 levels, IL1 levels, as well as by various polymorphisms in the CRP gene and promotor itself.

The ability to measure very low levels of CRP in "high-sensitivity CRP" or "hs-CRP" assays has allowed for further study of the role of C-reactive protein in chronic illness. It has been linked to cardiovascular disease, aortic valve stenosis, stroke, obesity, inflammatory bowel disease, and even to walking speeds and physical activity in the elderly. Chronic/low-grade inflammation and elevated CRP has also been associated with significantly higher risks of memory impairment.

Some of the CRP SNPs that are most strongly associated with negative clinical outcomes, such as cardiac disease or aortic valve stenosis, are associated with lower baseline CRP levels but appear to "over" respond to stimuli. This may be part of the pathogenesis of these inflammatory associations and risk of disease. Environmental factors and lifestyle choices such as diet, smoking, and exercise all can contribute to CRP and cytokine levels. In addition to genetics, baseline levels are also affected by weight, lipid levels, age, blood pressure, and of course inflammation and infection.

Inflammation Risk SNPs

Gene Name	CCL2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	8.9% of people have this result
Impact	High

"C-C Motif Chemokine Ligand 2" gene

CCL2 recruits more immune cells to sites of inflammation or infection. This can lead to more inflammation and tissue damage throughout your body.

- Variants are associated with an increased risk of:
 - Parkinson's in males
 - Knee Arthritis
 - Exercise-induced cardiac ischemia (low oxygen to the heart).
 - Diabetic foot ulcers
- Two variants are associated with an increased risk of:
 - Kidney disease with lupus (lupus nephritis)
 - Higher risk of heart disease if you have obstructive sleep apnea
 - Ischemic (decreased oxygenated blood supply to area of the brain) strokes
 - SARS-CoV (severe acute respiratory distress syndrome coronavirus)

Diet: A low glycemic index may be beneficial. You can ask your clinician about the glycemic handout found in the document library. Pomegranate helps lower CCL2.

Supplements: Palmitoylethanolamide (PEA), luteolin, pomegranate extract, sambucus, resolvins/SPMs, spirulina, astragalus, andrographis, vitamin A, and melatonin are supplements that can be considered with guidance on dosing, potential risks, and benefits.

Gene Name	TNF
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	31.3% of people have this result
Impact	Medium

"Tumor Necrosis Factor Alpha" gene

Tumor Necrosis Factor Alpha (TNF-alpha) is an inflammatory molecule that can cross the blood-brain barrier. This variant leads you to make more TNF-alpha, which can lead to inflammatory diseases, such as chronic gum disease.

- This variant is associated with an increased risk of a variety of conditions including:
 - Chronic gum disease
 - PCOS
 - Crohn's disease
- Two variants are associated with an increased risk of endometriosis.

Diet: Capsaicin (chili peppers) lowers TNF-alpha.

Supplements: Hericium erinaceus/Lion's mane, magnesium threonate, stinging nettle, sulforaphane, resolvins/SPMs, curcumin, berberine, EGCG, quercetin, rutin, silymarin (milk thistle), CBD, hydrogen water, choline, and betaine are supplements that can be considered with guidance on dosing, risks, and potential benefits.

Medications: Pioglitazone, carvedilol, and pentoxifylline are medications that have evidence of decreasing TNF-alpha that you can talk to your clinician about if a medication is indicated.

Gene Name	IL10
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	37% of people have this result
Impact	Medium

"Interleukin 10" gene

Interleukin 10 (IL10) is an important anti-inflammatory molecule. You may have lower levels of IL10, which makes you less able to "turn off" inflammation. This can predispose you to a variety of inflammatory conditions, such as chronic pain and asthma.

- This variant is associated with an increased risk of early pregnancy loss.
- Other inflammatory conditions that low IL10 levels are associated with include:
 - Chronic pain
 - Rheumatoid arthritis
 - Inflammatory Bowel Disease
 - Asthma
 - Psoriasis
- Two variants can also increase your risk of endometriosis.

Diet: Sesame oil, garlic, cinnamon, and mustard seed are foods that can help raise IL10.

Lifestyle: Exercise raises IL10.

Supplements: CBD, Bitter melon, quercetin, resolvins/SPMs, sulforaphane, curcumin, melatonin, bifidobacteria, and lactic acid bacteria blend (probiotics) are supplements that can be considered with guidance on dosing, risks, and potential benefits.

Medications: Estrogen and testosterone help raise IL10 levels. This can be discussed with your clinician about if you would benefit from either of them.

Gene Name	IL17A
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	38% of people have this result
Impact	Medium

"Interleukin 17A" gene

Interleukin 17A (IL17A) is an inflammatory molecule that plays a role in activating autoimmune pathways.

- This variant is associated with an increased risk of autoimmune diseases including:
 - Ulcerative colitis (type of inflammatory bowel disease)
 - Ankylosing Spondylitis
 - Graft versus host disease (body rejecting a transplant)
- Two variants are also associated with an increased risk of asthma, as well as an increased requirement for opioids after surgery.
- This variant may **decrease** your risk of having influenza A.

Supplements: Curcumin, grape seed extract, apigenin, fisetin, baicalin, CBD, palmitoylethanolamide (PEA), and sulforaphane are supplements that can be beneficial. Ask your clinician for guidance based on your gene variants but also symptoms, family history and your other medical conditions.

Gene Name	CRP
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	44.7% of people have this result
Impact	Medium

"C Reactive Protein" gene

CRP is a special kind of inflammation signal, and levels rapidly rise in response to blood vessel damage or infection. With this variant, when you have heart damage you tend to have increased levels of CRP. This does not relate to elevated CRP levels at your baseline.

- T allele associated with:
 - LOWER CRP levels normally, but increased serum CRP levels in the presence of aortic calcification and more severe aortic calcification (when there is blood vessel/heart damage).
 - Higher risk for severe community-acquired pneumonia.
- Men with two variants may have an increased risk of depression.

Diet: Ginger can help lower your CRP levels if they are high.

Supplements: Curcumin, CoQ10, magnesium, vitamin D, zinc, sulforaphane, probiotics, ginger, gamma tocotrienols, choline, and betaine are supplements that can help lower CRP levels if your levels are elevated. These can be considered with guidance on dosing.

Gene Name	IL1B
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	45.4% of people have this result
Impact	Medium

"Interleukin 1 Beta" gene

Interleukin 1 Beta (IL1B) is an inflammatory molecule that sends signals throughout the body. This variant leads you to release more IL1B which can be associated with an increased risk of many health issues like heart disease, and arthritis.

- This variant is associated with an increased risk of a variety of conditions including:
 - Heart disease in individuals who are 55 years of age or younger.
 - Febrile seizures
 - Arthritis
 - Autism
- Two variants are associated with an increased risk of septic shock, especially after having a recent surgery.
- Two variants also are associated with aspirin-associated respiratory disease.

Diet: Blueberries (source of pterostilbene), and red grapes (source of resveratrol) are foods that may be beneficial.

Supplements: Resolvins/SPM's, omega 3's, sulforaphane, pterostilbene, resveratrol, apigenin, fisetin, luteolin, and quercetin have evidence in decreasing IL1B and can be considered with guidance on dosing.

Medications: Carvedilol is a medication that has evidence of decreasing IL1B and can be discussed with your clinician if a medication is indicated.

Gene Name	NLRP3
Minor Allele	A
My Results	AC
My Variants	You have 1 minor allele
Prevalence	47.7% of people have this result
Impact	Medium

"NLR Family Pyrin Domain Containing 3" gene

NLRP3 is a protein that triggers the production of inflammasomes. Inflammasomes increase the production of many other inflammatory molecules. Two variants may lead to the overactivation of these inflammasomes which can increase the risk of damage to the lungs and other organs.

- Two variants are associated with an increased risk of coal worker's lung disease (pneumoconiosis). Smoking can further increase this risk.
- Two variants can also decrease your ability to respond to the treatment for hepatitis C, interferon.
- 0 variants are associated with a decreased risk of chronic hepatitis C infection.

Supplements: Resolvins/SPMs, sulforaphane, and berberine are supplements that can be considered with guidance on dosing.

Gene Name	CXCL8
Minor Allele	T
My Results	AT
My Variants	You have 1 minor allele
Prevalence	49.6% of people have this result
Impact	Medium

"C-X-C Motif Chemokine Ligand 8" gene

CXCL8 (also called IL8) guides immune cells towards areas of injury. This variant can lead you to have higher IL8 levels and can predispose you to changes in cognition and gout.

- This variant is associated with an increased risk of gout and cognitive changes.
- Two variants are associated with an increased risk of diabetic nerve disease.

Lifestyle: High-intensity interval panel (HIIT) can help decrease IL8.

Supplements: Quercetin, fisetin and Mitraphylline (Cat's Claw) are supplements that can be considered with guidance on dosing, risks, and potential benefits.

Magnesium

Magnesium is an essential nutrient that is involved in more than 300 enzyme systems that regulate protein synthesis, nerve function and blood glucose control. Although the exact role of magnesium in cognitive decline and Alzheimer's disease is still under investigation, serum and brain magnesium levels are decreased in individuals with cognitive impairment. Some signs of magnesium deficiency include irregular heartbeat, numbness, tingling, seizures and personality changes. Older adults tend to have low magnesium levels as they generally have lower magnesium diets, lower absorption of magnesium and excrete more magnesium in the kidneys.

Magnesium Risk SNPs

Gene Name	MECOM
Minor Allele	G
My Results	GA
My Variants	You have 1 minor allele
Prevalence	49% of people have this result
Impact	Medium

"MSD1 and EVI1 Complex Locus" gene

With 1 or 2 copies of this variant, you are prone to having low magnesium levels. This gene, but not necessarily this SNP, has been associated with aggression. Both high and low magnesium can contribute to mental/mood conditions, as well as aggression, through the influence magnesium has on brain chemicals like dopamine, adrenaline, and norepinephrine.

- This gene may be involved in a variety of cell cycle functions.
- Variants are associated with decreased magnesium levels.

Supplements: If low magnesium, supplementing with magnesium threonate or magnesium glycinate may be beneficial.

Diet: Increasing dietary intake of magnesium from foods can also support this variant. Ask your clinician to see the Magnesium Containing Foods Handout.

Obsession, Rumination, and Compulsions

Obsessive Compulsive Disorder (OCD) is a common, complex psychiatric disorder that is characterized by uncontrollable, recurring behaviors, known as "compulsions", and recurring thoughts, known as "obsessions". These traits can interfere with everyday life, work, school, and relationships.

There is growing evidence that some of the most significant risks in regard to OCD is genetic factors, particularly those that affect neurotransmitter levels in the brain. Neurotransmitters are the brain's chemical messengers responsible for sending signals between nerve cells in the brain. You will notice a lot of discussion on 4 brain chemicals: norepinephrine, dopamine, serotonin, and glutamate in this panel. Norepinephrine plays a role in energy, focus and motivation, dopamine plays a role in pleasure, movement and irritability and serotonin plays a role in mood, happiness and a sense of wellbeing. Glutamate, being the most abundant excitatory neurotransmitter, plays a role in learning, memory and cognition. With these chemical messengers the goal is the right balance - not too much and not too little while having an appropriate amount of each.

These neurotransmitters play a large role in the development of OCD, and can even cause resistance to OCD treatment. Lower serotonin levels, which can be due to genetic or dietary factors, such as low vitamin D, vitamin B12, and high homocysteine, have been correlated with OCD. Higher dopamine and glutamate levels, which are mainly due to genetic factors, have also shown association with OCD. Your licensed health care provider will discuss these variants with you and help you decide if you would benefit from any potential interventions and can help you determine which supplements and dietary changes make sense for you or your child. They also will determine appropriate dosing.

Obsession, Rumination, and Compulsions Risk SNPs

Gene Name	HTR2A
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	16.8% of people have this result
Impact	High

"5-Hydroxytryptamine Receptor 2A" gene

Individuals with this SNP are prone to less effective serotonin (happy, calming brain chemical) signaling. Low postsynaptic (receiving neuron) serotonin signaling is a major contributor to OCD.

- This variant leads to lower levels of serotonin.
- In addition to increasing the risk of OCD, this variant also increases the risk of "tic" disorder, and post-traumatic stress disorder (PTSD).

Diet: Foods that contain vitamin D (fish and fortified milk), foods that contain vitamin B6 (pistachios and chickpeas), foods that contain folate, (spinach and asparagus), and foods with vitamin B12 (clams and oysters) may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: Vitamin D, vitamin B6, vitamin B12, and folate are vitamins that may be beneficial. In addition, 5-HTP, L-theanine, certain probiotics, and myoinositol can be considered with guidance on dosing, potential risks, and benefits.

Medications: SSRIs, SNRIs, and TCAs are medications that help raise serotonin levels and can be considered by you and your clinician for treating OCD.

Gene Name	HTR2A
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	18.7% of people have this result
Impact	High

"5-Hydroxytryptamine Receptor 2A" gene

Individuals with this SNP are likely prone to less effective serotonin signaling. Serotonin is the "happy and calm" brain chemical (neurotransmitter). Low post-synaptic (receiving neuron) serotonin signaling is a major contributor to OCD.

- This variant increases the risk of OCD, and developing OCD at a younger age.
- This variant significantly increases the risk for "bedwetting" issues. Two variants (AA) has been associated with over 10 times the risk compared to just one variant having more than double the risk.
- The "non-variant", G allele, is associated with decreased likelihood to adopt the point of view of others, It is also associated with increased anxiety when observing the pain of others and increased likelihood of developing communication problems.

Diet: Foods that contain vitamin D (fish and fortified milk), foods that contain vitamin B6 (pistachios and chickpeas), foods that contain folate, (spinach and asparagus), and foods with vitamin B12 (clams and oysters) may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: Vitamin D, vitamin B6, vitamin B12, and folate are vitamins that may be beneficial. In addition, 5-HTP, L-theanine, certain probiotics and myoinositol can be considered with guidance on dosing, potential risks, and benefits.

Medications: SSRIs, SNRIs, and TCAs are medications that help raise serotonin levels and can be considered by you and your clinician for treating OCD.

Gene Name	HTR3C
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	19.3% of people have this result
Impact	High

"5-Hydroxytryptamine Receptor 3C" gene

Individuals with this SNP likely have less effective serotonin signaling due to changes in this serotonin receptor. Serotonin is the "calming and happy" brain chemical, and low serotonin signaling can contribute to OCD (this has been found to be significant in males).

- This variant has been associated with an increased risk of OCD in **males**.
- Individuals with this variant are more likely to have constipation-prone IBS type symptoms.

Diet: Foods that contain vitamin D (fish and fortified milk), foods that contain vitamin B6 (pistachios and chickpeas), foods that contain folate, (spinach and asparagus), and foods with vitamin B12 (clams and oysters) may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: Vitamin D, vitamin B6, vitamin B12, and folate are vitamins that may be beneficial. In addition, 5-HTP, L-theanine, certain probiotics, and myoinositol can be considered with guidance on dosing, potential risks, and benefits.

Medications: SSRIs, SNRIs, and TCAs are medications that help raise serotonin levels and can be considered by you and your clinician for treating OCD.

Gene Name	ANKK1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	32.1% of people have this result
Impact	Medium

"Ankyrin Repeat and Kinase Domain Containing 1" gene

Dopamine is a brain chemical/neurotransmitter involved in reward and motivation. Individuals with this SNP have increased dopamine signaling due to lower ability to regulate the dopamine pathway. This is associated with OCD.

- Too high of dopamine levels has been associated with Obsessive Compulsive Disorder (OCD), and avoidant and compulsive addictive behavior.
- This variant is also associated with an increased risk of ADHD, food addiction, and obesity.
- Individuals with this variant tend to have lower levels of BDNF. Think of BDNF as "fertilizer" for the brain. It is beneficial for memory and mental health.

Diet: You can ask your clinician about foods that can help raise serotonin (like foods rich in vitamin D and vitamin B6/B12), as serotonin can help inhibit dopamine. Foods rich in magnesium, such as pumpkin and soybeans, and zinc, such as oysters and spirulina, may also be beneficial. Avoiding a high sugar diet is also recommended.

Lifestyle: Exercise may be beneficial.

Supplements: Vitamins and minerals like vitamin D, vitamin B6, vitamin B12, magnesium, and zinc may be beneficial.

Medications: SSRIs are an example of medications that can raise serotonin if you have OCD.

Note: It may be beneficial to also look at the BDNF panel for a list of recommendations that can help raise BDNF levels.

Gene Name	PBX1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	34.7% of people have this result
Impact	Medium

"PBX Homeobox 1" gene

Individuals with this SNP are prone to greater dopamine (reward/motivation brain chemical) signaling due to more dopamine neurons (brain cells) being produced during development. Increased dopamine signaling lead to an increased OCD risk.

- This variant increases the risk of OCD, and is more significant in males than females.

Diet: You can ask your clinician about foods that can help raise serotonin (like foods rich in vitamin D and vitamin B6/B12), as serotonin can help inhibit dopamine. Foods rich in magnesium, such as pumpkin and soybeans, and zinc, such as oysters and spirulina, may also be beneficial. Avoiding a high sugar diet is also recommended.

Lifestyle: Exercise may be beneficial.

Supplements: Vitamins and minerals like vitamin D, vitamin B6, vitamin B12, magnesium, and zinc may be beneficial.

Medications: SSRIs are an example of medications that can raise serotonin if you have OCD.

Gene Name	SLC1A1
Minor Allele	C
My Results	GC
My Variants	You have 1 minor allele
Prevalence	44.1% of people have this result
Impact	Medium

"Solute Carrier Family 1 Member 1" gene

Individuals with this SNP are likely prone to greater glutamate levels. Glutamate is a brain chemical (neurotransmitter) that can irritate and overexcite the brain. Glutamate in excess can contribute to OCD and OCD symptoms such as nail biting, skin picking and hair pulling.

- The C allele has been associated with an increased risk for OCD. If combined with the gene variant above, there is a larger increased risk of OCD.
- This variant can increase the risk of treatment-resistant depression.

Diet: Foods that contain vitamin D, like fish and fortified milk, and foods with vitamin B12, such as clams and oysters, may be beneficial. You can ask your clinician for other recommendations for foods containing these vitamins. Eating a diet rich in selenium (for example eating 1-2 brazil nuts per day), can also be beneficial.

Supplements: NAC (N-acetylcysteine) can help lower glutamate. Vitamin D, vitamin B12, and 5HTP may also be of benefit and these supplements can be considered with guidance on dosing, potential risks, and benefits.

Gene Name	COMT V158M
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact	Medium

"Catechol-O-Methyltransferase" gene

Individuals with this SNP have a reduced ability to break down dopamine (motivation and reward brain chemical), norepinephrine, and adrenaline (fight or flight brain chemicals). High dopamine can encourage repetitive, reward-related behavior, and has been associated with OCD.

- This variant increases the risk of OCD, more so in men than women. It also increases the risk of anxiety.
- Two variants have been associated with:
 - More intolerance to caffeine.
 - Higher sensitivity to pain after surgery, and greater response to opiates when prescribed for pain.
 - **As a benefit** with motor learning skills.
- COMT is also involved in the breakdown of estrogen (hormone), so those with two copies may also have higher levels of estrogen.

Diet: Foods rich in magnesium, such as pumpkin and soybeans, and zinc, such as oysters and spirulina, may also be beneficial. Avoiding a sugar diet is also recommended.

Supplements: Ashwagandha can help lower cortisol levels. Zinc, magnesium, inositol, curcumin, and L-theanine may also be beneficial and can be considered with guidance on dosing.

Note: If you have two variants you may want to avoid quercetin, as it further reduces your ability to break down these brain chemicals.

Vitamin B12

Vitamin B12 is essential not only for the formation of red blood cells, but also for many other cellular functions. Vitamin B12 is used to clear homocysteine and is important for proper brain and neurological function. Low vitamin B12 is associated with neurological conditions such as Alzheimer's Disease and vascular dementia, and can also contribute to high homocysteine.

Seniors tend to have lower vitamin B12 levels due to problems with acid production and intrinsic factor, which is a molecule important in the absorption of vitamin B12. Vegetarians and vegans are at a high risk for B12 deficiency as almost all naturally occurring B12 is found in meat and dairy.

Cognitive health studies show higher cognitive function and less cognitive loss in individuals in the highest tertile of B12 levels. There are some B12 related variants that make it harder for tissue levels (TCN1 and TCN2) or mitochondrial levels (MUT) to stay normal even in the presence of normal blood B12 levels. Individuals with variants in this pathway may need higher levels of blood B12 in order to maintain adequate brain, mitochondrial and other tissue B12 levels.

Vitamin B12 Risk SNPs

Gene Name	TCN2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	19.8% of people have this result
Impact	High

"Transcobalamin 2" gene

Individuals with 2 copies of this SNP are prone to significantly lower levels of B12 in the brain. The carrier protein encoded by this gene only binds 10-20% of total B12, but is responsible for carrying 80-90% of the B12 into the brain. Individuals with this SNP may require higher levels of B12 in the blood to have adequate levels of B12 in the brain.

- This protein binds vitamin B12 and transports it into cells.
- GG variants have been associated with less efficient binding and transport of B12, correlating with reduced tissue levels of vitamin B12 (holotranscobalamin) even when B12 levels in the blood appear normal.
- GG variants have been associated with 3.33 times the risk of peripheral neuropathy even with normal B12 levels. Even greater risk (6.9 times) if folate intake is >800 mg/day.
- G allele has been associated with 1.51 times the risk for endometriosis.
- C allele (1 variant) has been shown to convey relative protection again Alzheimer's type dementia when combined with TT genotype of rs1801131 (MTHFR A129C).
- Individuals with this variant may need higher than average B12 levels in order to have adequate brain and tissue levels.

Supplements: Supplementing Vitamin B12 via injections, sublingual or oral forms can be beneficial to maintain adequate B12 levels. You may need higher doses to get enough B12 to the brain.

Diet: Increasing dietary intake of Vitamin B12 from foods such as meats, poultry, fish, etc. Ask your clinician for the Vitamin B12 Containing Foods handout.

Gene Name	CBS
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	20.6% of people have this result
Impact	High

"Cystathionine Beta Synthase" gene

Individuals with this SNP are prone to lower B12 and lower folate levels.

- Codes for enzyme that clears homocysteine using vitamin B6.
- CC has been associated with 1.22 times the risk of below adequate vitamin B12 status.
- CC has been associated with 1.56-2.5x times the risk for cleft palate.

Supplements: Supplementing Vitamin B12 via injections, sublingual or oral forms can be beneficial to maintain adequate B12 levels. Additionally, optimizing Vitamin B6 and folic acid.

Diet: Increasing dietary intake of Vitamin B12, B6, and folate from foods. Ask your clinician for the appropriate handouts.

Vitamin B12 Benefit SNPs

Gene Name	FUT2
Minor Allele	T
My Results	AT
My Variants	You have 1 minor allele
Prevalence	1.7% of people have this result
Impact	Helpful

"Fucosyltransferase 2" gene

This SNP results in making less of a protein that is normally found on the surface of blood cells where many bacteria and viruses bind. Having 2 copies of this SNP can be beneficial in conferring protection against infections such as H. Pylori and norovirus, but is also associated with producing less "good" bacteria in the gut and higher risk of Chron's disease. "Good" bacteria in the gut use up B12 for their own needs, so lower levels of these bacteria can result in higher levels of available B12.

- Codes for the secretion of blood type antigens (H antigens) into body fluids.
- Bacteria such as H. Pylori bind to H. antigens and interfere with release of intrinsic factor (found in the stomach), an essential compound for B12 absorption.
- Variant is associated with increased vitamin B12 levels.
- TT variants have been associated with 1.77 times the risk of Crohn's Disease.

Gene Name	ABCD4
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	46% of people have this result
Impact	Helpful

"ATP Binding Cassette Subfamily D Member 4" gene
This SNP contributes to higher B12 levels by converting B12 to its active form and transporting "activated" B12 into the cell where it can be used.

- Involved in the transport of B12 in and out of lysosomes (organelle in the cell that acts like a garbage collector).
- Variant is associated with elevated vitamin B12 levels.

Gene Name	FUT2
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	48.1% of people have this result
Impact	Helpful

"Fucosyltransferase 2" gene
This SNP results in making less of a protein that is normally found on the surface of blood cells where many bacteria and viruses bind. Having 2 copies of this SNP can be beneficial in conferring protection against infections such as H. Pylori and norovirus, but is also associated with producing less "good" bacteria in the gut and higher risk of Chron's disease. "Good" bacteria in the gut use up B12 for their own needs, so lower levels of these bacteria can result in higher levels of available B12.

- Codes for the secretion of blood type antigens (H antigens) into body fluids.
- Bacteria such as H. Pylori bind to H. antigens and interfere with release of intrinsic factor (found in the stomach), an essential compound for B12 absorption.
- "A" allele has been associated with increased vitamin B12 levels.
- "A" allele has been associated with 3.36 times the risk for Crohn's Disease or 1.29 times the risk for type 1 Diabetes Mellitus.

Vitamin B6

Vitamin B6 performs a variety of functions and is involved in cognitive development and clearing homocysteine. Elevated homocysteine levels have been associated with neurological conditions such as Alzheimer's Disease, vascular dementia and schizophrenia.

The ALPL gene produces an enzyme that clears vitamin B6. Variants in this gene are associated with lower B6 levels. B6 is used by a gene called CBS to clear homocysteine. Vitamin B6 supplementation has been shown to lower homocysteine. Discuss with your clinician potential interventions that are geared at raising B6 levels through diet and supplementation.

Common symptoms of low vitamin B6 levels include: fatigue, peripheral neuropathy, anemia, sore glossy tongue, mood changes, and cracked and sore lips.

Vitamin B6 Risk SNPs

Gene Name	NBPF3
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	22.9% of people have this result
Impact	High

"Neuroblastoma Breakpoint Family Member 3" gene

With this SNP, you are prone to removing your Vitamin B6 quicker than normal, due to this overactive pathway. This can lead to lower levels of vitamin B6.

- Located close to a gene involved in the clearance of vitamin B6.
- The variant T allele is associated with lower B6 levels.

Supplements: You may benefit from supplementation with Vitamin B6 or the active form P5P.

Diet: You can also increase your intake of dietary Vitamin B6 (see the Complete Vitamin B6 Containing Foods handout)

Note: It is important to remember that we want just the right amount of B6 - not too much or too little.

Gene Name	ALPL
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	30% of people have this result
Impact	High

"Alkaline Phosphatase, Liver/Bone/Kidney" gene

With this SNP, you are prone to removing your Vitamin B6 quicker than normal, due to this overactive pathway. This can lead to lower levels of vitamin B6.

- This protein is involved in the clearance of vitamin B6.
- Variant G allele is associated with lower B6 levels.

Supplements: You may benefit from supplementation with Vitamin B6 or the active form P5P.

Diet: You can also increase your intake of dietary Vitamin B6 (see the Complete Vitamin B6 Containing Foods handout)

Note: It is important to remember that we want just the right amount of B6 - not too much or too little.

Vitamin D

Vitamin D is a fat soluble vitamin that can be produced in the skin upon sunlight stimulation, or can be acquired via dietary/supplementary intake. Vitamin D is often referred to as a hormone due to the fact that it plays an integral role in hundreds of different pathways within the body. In addition to preventing bone-related disorders, adequate levels of vitamin D help maintain proper immune, neurological, muscular, and basic cellular function. Furthermore, recent research has shed light on the association between many inflammatory diseases and vitamin D deficiency. Unfortunately, a large percentage of the global population is either vitamin D deficient or insufficient. The form of vitamin D obtained from sunlight/dietary/supplementary intake is inactive and must undergo a 2-step conversion process to become the "active form" of vitamin D. The binding of activated vitamin D to its receptor ultimately results in regulation of hundreds of genes. Impaired regulation of these genes that are involved in various biological pathways can lead to a wide variety of undesirable health outcomes. Fortunately, correcting a vitamin D deficient status is fairly straightforward. Discuss with your clinician a strategy of initial correction, followed by a maintenance dose to bring your vitamin D levels into a healthy range.

Vitamin D Risk SNPs

Gene Name	VDR Apal
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	21.3% of people have this result
Impact	High

"Vitamin D Receptor" gene

This gene is a receptor that vitamin D binds to like a docking station. When bound to the receptor many genes can be either turned on or off. This variant is neither good nor bad. It may improve your bone health, make you taller and even lower the risk of asthma, it may increase your risk of polycystic ovaries.

- This variant is not associated with specific vitamin D levels.
- Variants are associated with higher bone mass/better bone structure, and also confer slight protection against asthma.

Note: This SNP is not necessarily one that HAS to be addressed. If concerned about autism speak with your clinician about ways to increase your vitamin D receptor expression

Supplements: Magnesium, sulforaphane, resveratrol, and genistein are some supplements that may be recommended and can be considered with guidance on proper dosing.

Gene Name	NADSYN1/
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	27.6% of people have this result
Impact	Medium

"7-Dehydrocholesterol Reductase" gene

Normally sunlight helps convert a building block of cholesterol to active vitamin D. With this variant, you have less ability to use the sun to convert cholesterol into vitamin D.

- Associated with lower vitamin D levels.
- This is not associated with higher cholesterol numbers.

It may be beneficial for you to get your vitamin D levels checked.

Diet: Incorporating vitamin D into your diet is beneficial. Examples of vitamin D-rich foods include maitake mushrooms, salmon, and tuna.

Supplements: Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.

Gene Name	CYP24A1
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	37% of people have this result
Impact	

"Cytochrome P450 Family 24 Subfamily A Member 1" gene

You have increased activity of an enzyme that breaks down vitamin D.

- This variant is associated with lower vitamin D levels.

It may be beneficial for you to get your vitamin D levels checked.

Diet: Incorporating vitamin D into your diet is beneficial. Examples of vitamin D-rich foods include maitake mushrooms, salmon, and tuna.

Supplements: Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.

Gene Name	GC
Minor Allele	G
My Results	TG
My Variants	You have 1 minor allele
Prevalence	39.8% of people have this result
Impact	

"Vitamin D Binding Protein" gene

Vitamin D binding protein is the main carrier for vitamin D. This variant decreases the amount of vitamin D binding protein concentrations, leading to decreased transport of vitamin D around your body.

- This is the **most significant SNP** that relates to vitamin D levels (and needs). Individuals with 2 variants in this pathway **may need as much as 40,000 to 50,000 units of vitamin D weekly** in order to maintain levels. Consult with your clinician.

It may be beneficial for you to get your vitamin D levels checked.

Diet: Incorporating vitamin D into your diet is beneficial. Examples of vitamin D-rich foods include maitake mushrooms, salmon, and tuna.

Supplements: Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.

Note: If you have 2 copies of this variant you may need a much higher dose of vitamin D than typical.

Gene Name	DHCR7
Minor Allele	G
My Results	GT
My Variants	You have 1 minor allele
Prevalence	40.1% of people have this result
Impact	Medium

"7-Dehydrocholesterol Reductase" gene

Normally sunlight helps convert a building block of cholesterol to active vitamin D. With this variant, you have less ability to use the sun to convert cholesterol into vitamin D.

- Associated with lower levels of vitamin D.
- This is not associated with higher cholesterol levels.

It may be beneficial for you to get your vitamin D levels checked.

Diet: Incorporating vitamin D into your diet is beneficial. Examples of vitamin D-rich foods include maitake mushrooms, salmon, and tuna.

Supplements: Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.

Vitamin D Benefit SNPs

Gene Name	CYP2R1
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	46.4% of people have this result
Impact	Helpful

"Cytochrome P450 Family 2 Subfamily R Member 1" gene

If you have two variants you have increased activity of an enzyme that helps convert vitamin D to its active form.

- Note that **having 1 variant is actually a risk** for vitamin D deficiency. 2 variants confers a relative increased ability to convert vitamin D to the storage form.

1 (or 0 variant): It may be beneficial to get your vitamin D levels checked. Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.

Gene Name	GC
Minor Allele	C
My Results	AC
My Variants	You have 1 minor allele
Prevalence	49% of people have this result
Impact	Helpful

"Vitamin D Binding Protein" gene

Vitamin D binding protein is responsible for binding and transporting vitamin D in your body. If you have two variants you likely have a better ability to transport vitamin D throughout your body.

- **Having 1 variant is actually a risk** and is associated with lower vitamin D levels. Individuals with 2 variants are relatively less likely to have impaired vitamin D transport.

***1 (or 0) variant:** It may be beneficial to get your vitamin D levels checked. Magnesium helps convert vitamin D to its active form and can be considered in combination with a vitamin D supplement with guidance on proper dosing.*

Viral Induced Lung Inflammation and ARDS

This panel focuses on some of the top variants associated with viral-induced lung inflammation and Acute Respiratory Distress Syndrome (ARDS). This has become a growing concern due to the SARS-CoV-2 pandemic. This panel is temporarily available as part of all our IntellxxDNA reports at no additional cost. Our hope is that it will serve as a discussion point between you and your clinicians, allowing all to better understand some of the root causes of viral-induced lung inflammation and ARDS. Please note, as in accordance with FTC/FDA regulations, we are not making coronavirus claims for any specific products, vitamins or supplements. As a clinical decision support tool, we are providing your clinician with evidence-based information to help support you.

Novel COVID19 is HIGHLY contagious and everyone should see themselves as "at risk" for COVID-19. That said, there are certain gene variants that are useful to be aware of because they can lead to an increased risk of infection, as well as severe infections, including ARDS. This information is also important for individuals who have chronic health conditions associated with increased risk of ARDS, such as severe obesity. With the panel below, we will provide some interventions that you can speak to your clinician about to help prevent a severe case of SARS-CoV-2 based on your genomics.

IntellxxDNA™ is not a risk calculator. Risk of COVID and or ARDS, as well as risk of all inflammatory, infectious diseases, chronic illnesses is multi-factorial with factors such as obesity, nutrient levels, and the effects of many many genes needing to be considered. Again, everyone should consider themselves at high risk for getting coronavirus.

NOTE: This information is not meant to imply that individuals with specific SNPs are at higher risk for infection or adverse events from COVID19 as studies have not yet been done on this specific virus with the specific genomic SNPs listed. IntellxxDNA is a clinical decision support tool that can help licensed health care providers make better clinical decisions and should be used along with history, labs and physical exam. By listing potential interventions for specific SNPs we are not making coronavirus treatment claims, but just giving further information on mechanism of action of various supplements, foods and vitamins in the published literature.

Viral Induced Lung Inflammation and ARDS Risk SNPs

Gene Name	ABO
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	5.1% of people have this result
Impact	High

"ABO Blood Group" gene

This gene corresponds to a non-O blood type. Non-O blood types may be at higher risk of SARS-CoV-2 infections.

- Variants correlate with non-O blood type.
- O blood type has been associated with lower risk of SARS-CoV-2 infections.
- A blood type has been associated with greater risk of SARS-CoV-2 infections.

Supplements: Curcumin, pycnogenol, nattokinase, bromelain, rutin, luteolin, quercetin, nattokinase, and lumbrokinase are supplements that can decrease platelet clumping (aggregation), and can be considered with guidance on dosing.

Medication: Depending on your other risk factors your clinician may recommend aspirin with this variant.

Gene Name	CCL2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	8.9% of people have this result
Impact	High

"C-C Motif Chemokine Ligand 2" gene

You may get more tissue damage from inflammation or infections. Two copies of this variant are associated with an increased risk of severe acute respiratory syndrome coronavirus (SARS-CoV) infection.

- Molecule that helps guide immune cells towards sites of inflammation and tissue injury.
- Studies show that CCL2 is increased during infection and is correlated with lung inflammation and damage.
- Having 2 variants also associated with increased risk of severe acute respiratory syndrome coronavirus (SARS-CoV) based on study done on the original SARS-CoV.

Supplements: Sambucus, resolvins/SPMs, astragalus, andrographis, luteolin, pomegranate, spirulina, vitamin A, and melatonin are some supplements that can help lower this type of inflammation and can be considered with guidance on dosing.

Gene Name	NLRP3
Minor Allele	A
My Results	AC
My Variants	You have 1 minor allele
Prevalence	47.7% of people have this result
Impact	Medium

"NLR Family Pyrin Domain Containing 3" gene

You may be more likely to activate something called inflammasomes in your body. SARS-CoV also activates inflammasomes. Too much inflammation can lead to an increased risk of lung damage from infection.

- Involved in the inflammatory pathway and activation of inflammasomes (which are mediators of inflammation in our body). Inflammasomes and inflammation can be triggered by things like viruses. Individuals with 2 SNPs in this pathway will tend towards increased production of these inflammasomes and thus inflammation.
- Having 2 variants has been associated with increased risk of pneumoconiosis (lung inflammatory reaction). Smoking can increase risk.

Supplements: Resolvins/SPMs and sulforaphane inhibit inflammasomes. Other supplements that can decrease inflammasomes, and be considered with guidance on dosing, include quercetin, melatonin, andrographis, astragalus, and vitamin C.

Endocannabinoid

The cannabinoid system is not just about THC and marijuana response. The endocannabinoid system (ECS) is a crucial signaling system that regulates many different aspects of the body such as memory and learning, gastrointestinal motility, appetite, energy regulation, immune response, substance addiction, and mood. The endocannabinoid system is also involved in brain development, reward, and stress response. Furthermore, disturbances in one's ECS system is associated with various disorders including Autism spectrum disorder (ASD), cognitive impairment, depression, anxiety, and schizophrenia.

This pathway is composed of two receptors - cannabinoid receptor 1 (CB1) and cannabinoid receptor 2 (CB2). CB1 receptors are mainly located in the brain and central nervous system (highly expressed on nerve cells). CB2 receptors are primarily expressed outside of the brain and are more involved in the immune system, but they are also found in the central nervous system (brain stem and microglial cells) under neuroinflammatory conditions (such as Alzheimer's and ALS). This panel looks at small changes in DNA called SNP's or variants that may influence CB1 and CB2 signalling.

The pathway is also comprised of two main endocannabinoids: Anandamide (AEA) and 2-arachidonoylglycerol (2-AG). These are the most heavily studied endocannabinoids and can be synthesized in the body from lipid precursors. Anandamide is referred to as the "bliss molecule" (produced by the body during meditation, etc and also part of the "runners high" effect). 2-AG has been shown to have many positive health effects decrease inflammation, mitigate seizure activity, etc.

Endocannabinoid Risk SNPs

Gene Name	CNR1
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	5.9% of people have this result
Impact	High

"Cannabinoid Receptor 1" gene

Individuals with this SNP have altered endocannabinoid signaling. This can increase risk for cannabis dependence, and decrease risk for dependence on other drugs such as cocaine, opioids, and alcohol.

- Variant has been associated with decreased expression of CNR1 receptor
- Variant has been associated with predisposition to cannabis and other substances such as nicotine and alcohol.

Diet: Foods that are high in Omega 3s can be beneficial, as well as dark chocolate and cacao. Cooking with olive oil may provide benefits due to a chemical, called hydroxytyrosol.

Supplements: Palmitoylethanolamide (PEA), hydroxytyrosol (from olive oil), and N-acetylcysteine (NAC) can be beneficial for this variant.

Gene Name	FAAH
Minor Allele	A
My Results	CA
My Variants	You have 1 minor allele
Prevalence	32.4% of people have this result
Impact	Medium

"Fatty Acid Amide Hydrolase" gene

Individuals with this SNP have decreased FAAH expression. This may lead to greater sensitivity to CBD, as well as increased appetite and pain sensitivity.

- You have decreased ability to break down anandamide.
- Has been associated with increased pain sensitivity.
- Two variants has been shown to have increased risk of alcohol and cannabis dependence.

Gene Name	FAAH
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	46.1% of people have this result
Impact	Medium

"Fatty Acid Amide Hydrolase" gene
Individuals with this SNP likely have decreased FAAH activity. This may lead to greater sensitivity to CBD/THC, and increase risk of psychosis when using cannabis.

- This variant has been shown to reduce the ability to break down the byproducts of CBD.
- You have a significantly increased risk for THC/CBD-induced anxiety.
- It may be beneficial for you to start on a lower dose than normal for a CBD regimen.

Endocannabinoid Benefit SNPs

Gene Name	ABCB1
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	49.1% of people have this result
Impact	Helpful

"ATP Binding Cassette Subfamily B Member 1" gene
Individuals with this SNP have decreased ability to move THC out of the brain, increasing risk for cannabis induced psychosis. This may result in lower risk of cannabis dependence.

- This variant is associated with a decreased risk of cannabis dependence.
- The AA genotype is also associated with better remission rates if using certain antidepressants.

Addiction and Substance Use Disorder

There are different brain chemicals that can get out of balance and contribute to drug use and addiction. Your genes play a significant role in addiction and substance use disorder. These genetic factors are estimated to contribute 40–60% to the total risk of addiction and drug dependency. Environmental factors, such as who you socialize with, and your job can also contribute to drug abuse and addiction. The good news is that when you understand some of your genetic contributing factors you can better address them!

This panel discusses a wide variety of types of substance abuse and addiction including nicotine, alcohol, and opioids. There are a few genes in this panel related to cannabis addiction, but there is also a whole separate section relating to the "endocannabinoid" system which has many gene variants that relate to cannabis or marijuana. A lot of addiction relates to either high dopamine (which also contributes to things like ADHD and even OCD), or high glutamate (which also contributes to OCD), and to changes in one's opioid receptors.

The panel looks at slight changes in DNA called SNPs or variants. An example of how genetic variants, can lead to addiction relates to the brain chemical dopamine. Too high of dopamine (responsible for the sensation of reward) or too low of dopamine tends to trigger addiction. Why? A key part of addiction is that our brains are programmed to desire the release of dopamine since it is the "reward" brain chemical. However, too much dopamine can be bad for your brain cells. To protect our brains, when dopamine gets too high, the brain responds by decreasing the number of dopamine receptors (binding sites). Fewer dopamine receptors then trigger a "dopamine reward deficiency syndrome" and can lead to people craving things, like alcohol or opiates. These substances often can increase dopamine release in the brain. If this happens, with less dopamine receptors, an individual may constantly need or crave more of the substance to get the same effect and the same amount of dopamine. This creates a "dopamine loop". Understanding the root cause is helpful to allow clinicians to treat people with addiction individually based on their genetics. For example, someone with high glutamate may benefit from N-acetylcysteine (NAC), which is a supplement that has been shown to help in addiction and lowers glutamate, while someone with high dopamine may benefit from vagus nerve stimulation which lowers dopamine. In addition to supplements, speak to your clinician if you would like to know what medication you would benefit from to help quit things like nicotine, or alcohol. There are certain medications you may benefit from based on your genomics.

Addiction and Substance Use Disorder Risk SNPs

Gene Name	OPRM1
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	2.8% of people have this result
Impact	High

"Opioid Receptor Mu 1" gene

You may have an increased risk of opioid addiction, as well as addiction in general. This gene variant gives you a better ability to respond to addiction treatments that act on the opioid receptor, including the medication Naltrexone.

- This gene variant may increase your risk of becoming addicted to things like alcohol.
- You may have a higher risk of having a cardiac arrest from overdosing from drugs like opioids (heroin is an example), or benzodiazepenes (Xanax is an example).
- Two variants have been associated with psychosis after treatment with ADHD medications with methamphetamines in them.

Supplements: Magnesium, zinc, omega 3's, chamomile, and ashwagandha are supplements that you may benefit from and can be considered with guidance on dosing.

Medications: Naltrexone is a medication that can be considered for treating addiction. If you have ADHD and two variants you may want to avoid stimulants with methamphetamines in them.

Gene Name	DRD2
Minor Allele	A
My Results	CA
My Variants	You have 1 minor allele
Prevalence	25.7% of people have this result
Impact	Medium

"Dopamine Receptor D2" gene

You may have less ability to regulate dopamine (a brain chemical) leading to more of the dopamine being released in your brain. Dopamine is responsible for making you feel rewarded for things and has specifically been linked to cocaine addiction.

- This variant is commonly found in individuals addicted to **cocaine**. It is also associated with an increased risk of death related to cocaine use.
- "A" allele is associated with better positive learning, as well as worse negative learning. This means that if you have this variant you may learn better when you are given positive feedback on what you are learning.
- This variant is associated with a **decreased risk** of binge eating disorder.

Diet: Consider consuming foods high in butyrate (asparagus, garlic, onions, leeks, etc.). You can ask your clinician for a handout that contains a further list of foods that contain butyrate.

Lifestyle: Counseling may be encouraged due to the higher sensitivity to substances such as alcohol, opioids. Exercise and sunshine may also help. There are vagal nerve stimulators (wrist devices or Neuvana), that can increase "vagal tone", which has been shown to help addiction.

Supplements: Citicoline, resveratrol, ginseng, and magnesium all have evidence for helping with cocaine addiction and related effects and can be considered with guidance on dosing.

Medications: Disulfiram may be beneficial for the treatment of cocaine addiction.

Gene Name	DRD2
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	26.4% of people have this result
Impact	Medium

"Dopamine Receptor D2" gene

This gene change makes your body less able to put the "brakes" on dopamine. Dopamine is a brain chemical that makes you feel good and rewarded for things. Too much dopamine can trigger addictive behaviors.

- This variant is associated with a higher risk of alcohol, cocaine, and nicotine dependence.

Diet Eating foods that are high in butyrate (asparagus, garlic, onions, leeks, etc.) may be beneficial. You can ask your clinician for a handout that contains a further list of foods that contain butyrate.

Lifestyle: Sunshine and exercising may be beneficial. There are vagal nerve stimulators (wrist devices or Neuvana), that can increase "vagal tone", which has been shown to help addiction.

Supplements: Citicoline, bacopa monnieri, and berberine are supplements that can be considered with guidance on dosing.

Gene Name	ANKK1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	32.1% of people have this result
Impact	Medium

"Ankyrin Repeat and Kinase Domain Containing 1" gene

This gene normally allows you to regulate dopamine levels, kind of like a gas gauge. With this variant your "regulator" is off, which can lead to more dopamine in your brain. Dopamine is an important part of your reward system, so you may have a bigger "reward response" which can trigger addictive behaviors relating to everything from food and alcohol to video games and narcotics.

- You may have a higher risk for addictive behavior with regards to substances and medications (narcotics), but also food, alcohol, nicotine, and video games.
- This also relates to an increased risk of ADHD.
- If you are male and have two copies of this variant you may be at increased risk of having depression.

Diet: Eating foods that are high in butyrate (asparagus, garlic, onions, leeks, etc.) may be beneficial. You can ask your clinician for a handout that contains a further list of foods that contain butyrate.

Lifestyle: Sunshine and exercise may be beneficial. There are vagal nerve stimulators (wrist devices or Neuvana), that can increase "vagal tone", which has been shown to help addiction.

Supplements: Citicoline, NAD, bacopa monnieri, and berberine may be beneficial for you and can be considered with guidance on dosing.

Gene Name	FAAH
Minor Allele	A
My Results	CA
My Variants	You have 1 minor allele
Prevalence	32.4% of people have this result
Impact	Medium

"Fatty Acid Amide Hydrolase" gene

You may have more of a bliss molecular called anandamide, which is an endocannabinoid. This can also increase dopamine levels (which makes you feel rewarded). You may feel more rewarded from alcohol and some drugs; however, you may have a decreased risk of marijuana abuse. This is important to consider if you have two copies of this variant.

- Having this variant may decrease your ability to inactivate a bliss molecular in your brain that may lead to increased reward sensation.
- Two copies of this variant are associated with 3-5 times the risk for alcohol and drug addiction. This variant is also associated with an increased risk of methamphetamine (stimulant) dependence.
- 2 copies of this variant may decrease your risk of marijuana dependence.
- Interesting fact: Exercise and chocolate increase the bliss molecule (anandamide). Individuals with variants already have high levels of this molecule.

Supplements: Resveratrol (from grapes), PEA, NAC, genistein, and omega 3's such as DHA and EPA are supplements that may be beneficial and can be considered with guidance on dosing.

Gene Name	TPH1
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	47.5% of people have this result
Impact	Medium

"Tryptophan Hydroxylase 1" gene

You may become more constipated after receiving opiates if you have two copies of this variant.

- In a study of cancer patients receiving opioids (for pain relief), people with two variants reported more constipation than those without two variants.

Gene Name	GABRA2
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	47.6% of people have this result
Impact	Medium

"GABA Type A Receptor Subunit Alpha 2" gene

You may have less response to GABA, which is the "calming" brain chemical. This can decrease your ability to stop your brain reward center.

- This variant has been associated with a greater risk of aggressive behavior after drinking alcohol and alcohol dependence (relying on alcohol).
- It is also associated with increased marijuana, heroin, and opioid abuse.

Supplements: L-theanine, ashwagandha, resveratrol, and bacopa monnieri may be beneficial and can be considered with guidance on dosing.

Medications: Medications like Campral (acamprosate), naltrexone, or topiramate can help with addiction.

Gene Name	ABCB1
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	49.1% of people have this result
Impact	Medium

"ATP Binding Cassette Subfamily B Member 1" gene

You may have difficulty with clearing toxins and drugs out of the brain as well as someone who does not have this gene variant. This can lead to higher amounts of drugs staying in the brain.

- You may have more sensitivity to opioids such as fentanyl and morphine and so you may need lower doses when prescribed for pain.
- If you have two copies of this variant, you may have an increased risk of having psychotic effects on your brain after marijuana use.
- There is also evidence that this variant increases the risk of memory changes.

Lifestyle: Maintaining a healthy BMI is beneficial.

Supplements: PEA, CBD, and NAC are examples of supplements that may help decrease marijuana dependence and can be considered with guidance on dosing, risks, and potential benefits.

Medications: You may need a lower dose of opioid/pain medications if you are prescribed them. You can discuss this with your clinician.

Gene Name	TTC12
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	47.6% of people have this result
Impact	

"Tetratricopeptide Repeat Domain 12" gene

If you have two variants you are less likely to smoke because you may feel less rewarded from nicotine.

- This is a benefit only if you have two variants.
- Having one or zero variants is associated with an increased risk of smoking (both occasionally and regularly), with stronger effects noted in adolescent individuals.

If 1 (or 0) variants and difficulty with smoking cessation: L-theanine and Rhodiola rosea are supplements that may help. Also, ask your clinician about foods that contain tryptophan that may be beneficial. Speak with your clinician before starting any supplements and ask about the appropriate dosing of supplements that your clinician recommends for you.

BDNF

The BDNF gene codes for a brain growth factor called brain-derived neurotrophic factor, which is essentially fertilizer for the "neurons" or brain cells. BDNF is important in the brain as it plays a crucial role in not only protecting nerve and brain cells but also in growth and maturation. This nerve growth factor is important for learning and memory via its role in strengthening brain connections with use. Nerve connections can weaken when not used. This is similar to what we see with other tissues such as muscles; You use it or lose it. Individuals with lower levels of BDNF have more difficulty stimulating brain growth, and therefore BDNF SNPs have been associated with Alzheimer's, autism, cognitive decline, depression, and other neurological conditions. Studies have shown that BDNF levels are lowered with depression and normalize with remission. this panel looks at slight changes in DNA called SNP's or variants that can impact your BDNF levels.

Moderate to intense aerobic exercise, as well as supplements such as butyrate and curcumin have been shown to raise BDNF levels. There are also many gut bacteria that can produce butyrate, but since most of them are "anaerobes" (live where there is no oxygen), these species do not survive well in bottles of probiotics. However, keeping a diet high in prebiotic high fiber foods such as high fiber vegetables like asparagus, onions, whole grains and inulin (jerusalem artichokes) can help increase the butyrate producing bacteria in the gut.

BDNF Risk SNPs

Gene Name	BDNF
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	31.1% of people have this result
Impact	Medium

"Brain-Derived Neurotrophic Factor" gene

Brain-derived neurotrophic factor (BDNF) is a growth factor, essentially "fertilizer" for the brain. It is especially important for memory and mental health. You may not be able to convert the pre form of BDNF called pro-BDNF to the mature activate BDNF. Only the mature form helps nerve cells to grow.

- This variant is associated with an increased risk of cognitive impairment or Alzheimer's. This risk is compounded if in combination with variants in each of the two SNP's below (rs11030104, and rs2049045). When all 3 of the SNPs are present risk can be 2.7x higher. This can be overcome with intense exercise as discussed below.
- This has also been associated with an increased risk for more severe depression and dry eye disease.
- You may have a better response to certain antidepressant medications (SNRI's like duloxetine and venlafaxine and TCAs like nortriptyline and amitriptyline); however, you are less likely to respond to SSRIs like citalopram, escitalopram, fluoxetine and sertraline.

Diet: Consuming butyrate-producing foods, as well as high fiber foods, blueberries and prebiotics may also help raise these levels. You can ask about obtaining the Butyrate Rich Foods handout found in the Document Library.

Lifestyle: Moderate to intense exercise (generally 30-45 minutes multiple days a week) .**This has some of the best evidence in raising BDNF levels.

Supplements: Curcumin, lutein, zeaxanthin, certain probiotics, and silymarin are examples of supplements that can help raise BDNF levels and can be considered with guidance on dosing, risks, and potential benefits.

Gene Name	BDNF
Minor Allele	G
My Results	AG
My Variants	You have 1 minor allele
Prevalence	33.5% of people have this result
Impact	Medium

"Brain-Derived Neurotrophic Factor" gene

Brain-derived neurotrophic factor (BDNF) is a growth factor, essentially "fertilizer" for the brain. It is especially important for memory and mental health. You may make less of this BDNF growth factor.

- This variant combined with variants in rs6265 above, and rs2049045 below may increase your risk of cognitive impairment.

Diet: Consuming butyrate-producing foods, as well as high fiber foods, blueberries and prebiotics may also help raise these levels. You can ask about obtaining the Butyrate Rich Foods handout found in the Document Library.

Lifestyle: Moderate to intense exercise (generally 30-45 minutes multiple days a week) . **This has some of the best evidence in raising BDNF levels.

Supplements: Curcumin, lutein, zeaxanthin, certain probiotics, lion's mane, and silymarin are examples of supplements that can help raise BDNF levels and can be considered with guidance on dosing, risks, and potential benefits.

Gene Name	BDNF
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	48% of people have this result
Impact	Medium

"Brain-Derived Neurotrophic Factor" gene

Brain-derived neurotrophic factor (BDNF) is a growth factor, kind of like fertilizer for the brain. This is a more minor BDNF SNP, but relates to the response of anti-depressants.

- You may have a higher rate of brain volume reduction (left hippocampus), which can lead to increased memory loss.
- This is also associated with an increased risk of Alzheimer's related depression.
- There is some evidence that shows that individuals with this variant may respond better to antidepressants.

Diet: Consuming butyrate-producing foods, as well as high fiber foods, blueberries and prebiotics may also help raise these levels. You can ask about obtaining the Butyrate Rich Foods handout found in the Document Library.

Lifestyle: Moderate to intense exercise (generally 30-45 minutes multiple days a week) . **This has some of the best evidence in raising BDNF levels.

Supplements: Curcumin, lutein, zeaxanthin, certain probiotics, lion's mane, and silymarin are examples of supplements that can help raise BDNF levels and can be considered with guidance on dosing, risks, and potential benefits.

Gene Name	BDNF
Minor Allele	G
My Results	AG
My Variants	You have 1 minor allele
Prevalence	48.9% of people have this result
Impact	 Medium

"Brain-Derived Neurotrophic Factor" gene

Brain-derived neurotrophic factor (BDNF) is a growth factor, kind of like fertilizer for the brain. While the exact mechanism of how this variant affects BDNF is unclear, this variant impacts verbal learning.

- This variant may lead to you scoring lower on a learning test that is used as a marker for the development of cognitive impairment.
- There has been an association with decreased executive functioning skills in children who have ADHD (self-control, planning, organization, etc.).

Diet: Consuming butyrate-producing foods, as well as high fiber foods, blueberries and prebiotics may also help raise these levels. You can ask about obtaining the Butyrate Rich Foods handout found in the Document Library.

Lifestyle: Moderate to intense exercise (generally 30-45 minutes multiple days a week) .**This has some of the best evidence in raising BDNF levels.
Supplements: Curcumin, lutein, zeaxanthin, certain probiotics, and silymarin are examples of supplements that can help raise BDNF levels and can be considered with guidance on dosing, risks, and potential benefits.

Environmental Toxins (Benzene)

The pathways reviewed in this panel give important information about detoxification pathways needed to clear toxins from smoke, benzene, and organophosphate-containing pesticides (found on many non-organic forms of produce and coffee beans). NOTE: there are other pathways in the mercury detox/glutathione detoxification which are also very important for clearing various pesticides with the most famous being glyphosate (commercially sold as "Round-Up"). Therefore, the Mercury Detox panel should also be carefully considered when discussing toxin exposure and clearance.

Benzene exposure is common today and has been associated with poor cognition. Common sources of benzene exposure include gasoline and traffic-related air pollution, cigarette smoking and secondhand smoke, glues, solvents, and paints. Benzene has known cancer-causing properties. Variants in some genes listed below (NQO1, CYP2E1, and EPHX1) change how our body processes benzene, and can lead to toxicity. Organic phosphates (OPs) are found in pesticides. Non-organic fruits and vegetables, as well as non-organic coffee beans, are common sources of OPs. The PON1 SNP included in this panel is the most significant PON1 variant correlating with decreased ability to clear organic pesticides. Because infants and children already have less PON1 activity, this variant is even more significant in babies and young children. There are studies that show that you can decrease your pesticide exposure to about 80% by eating organic for the twelve foods that have the most pesticides. These foods are frequently called the "dirty dozen". See potential interventions for a list of the "dirty dozen" as well as the "clean 15" (foods that have fewer pesticides on them).

Environmental Toxins (Benzene) Risk SNPs

Gene Name	ALDH2
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	0.8% of people have this result
Impact	High

"Aldehyde Dehydrogenase 2" gene

Your body may have difficulty with processing alcohol, which can lead to a greater build-up of the toxin formaldehyde in the blood and brain. Higher levels of formaldehyde can harm memory and your heart.

- This variant may lead to you having alcohol "flushing" after consuming alcohol.
- This variant also relates to an increased risk of developing problems with cognition, heart disease, and early pregnancy loss.

Diet: Incorporating asparagus, Korean pear (Asian pear), mango, and ginseng into your diet may be beneficial.

Lifestyle: It is best to avoid alcohol consumption.

Supplements: Sulforaphane, alpha-lipoic acid, ginseng, taurine, pantetheine, and NAD+ are examples of supplements that you may benefit from and can be considered with guidance on dosing.

Gene Name	ADH1B
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	1.8% of people have this result
Impact	High

"Alcohol Dehydrogenase 1B" gene

This makes you convert alcohol to a more toxic intermediate compound up to 80 times faster than individuals with no variants. As a result, this can lead you to become hyper-sensitive to alcohol, have decreased cognition, and feel worse after drinking alcohol.

- You may drink less than people without this variant and have decreased risk of becoming dependent on alcohol.
- This variant may also increase your risk of restless leg syndrome as well as problems with cognition.

Diet: Incorporating asparagus, Korean pear (Asian pear), mango, and ginseng into your diet may be beneficial.

Lifestyle: It is best to avoid alcohol consumption. Incorporating asparagus, and ginseng into your diet may also be beneficial.

Supplements: NAD+ and ginseng are supplements that may be considered with guidance on dosing.

Gene Name	NQO1
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	4.1% of people have this result
Impact	High

"NAD(P)H Quinone Dehydrogenase 1" gene

This gene codes for an enzyme that is responsible for converting CoQ10 and vitamin E to their active forms. Two copies of this gene decrease this enzyme's activity by as much as 96-98%. This leads to a decreased ability to get rid of toxins, especially things like benzene and smoke.

- This variant is associated with a significant decrease in the activity of this enzyme, even more so if you have two variants.
- Decrease activity of this enzyme increases the risk of benzene poisoning/liver toxicity if you are exposed to air pollution.
- Cigarette smoke, gasoline, and plastic water bottles are examples of sources of benzene.

Lifestyle: Avoiding cigarette smoke, and limiting plastic water bottle use are examples of limiting benzene exposure that can be beneficial if you have this variant.

Supplements: Sulforaphane and Carnosic acid/carnosol (from Rosemary) help increase the activity of this enzyme. CoQ10, vitamin E, hydrogen water, glutathione, and NAD+ can also be considered with guidance on dosing, risks, and potential benefits.

Heavy Metals Detox & Glutathione

This panel focuses on genes and variants involved in mercury (Hg) toxicity, but these same variants play important roles in our master detoxification system needed for removing not only mercury and other metals such as arsenic, but also many other toxins. The most clinically significant SNPs in this panel are glutathione related. The glutathione pathway is considered the master detoxification pathway that removes toxins and carcinogens from various tissues within the body. Glutathione is particularly important for brain health. Glutathione SNPs have been implicated in over 200 different health and disease states.

Mercury toxicity is a known cause of cognitive impairment. Neurological symptoms of mercury toxicity include: depression, irritability, memory loss, insomnia, fatigue, and even hand tremors. Many cardiovascular conditions can also stem from chronic Hg exposure. Common sources of mercury exposure include environmental/chronic low level exposures such as frequent consumption of the larger predatory fish like swordfish and tuna, occupational exposure in which inhalation of mercury vapor is common, and dental amalgam fillings in which 50% of the filling is mercury.

There is an interesting relationship between mercury and the APOE SNPs. Individuals carrying APOE e4 are less able to protect against mercury damage compared to APOE e2 and e3. This is because the APOE e4 individuals are missing the piece of the lipoprotein that efficiently binds the mercury and helps with clearing it. Much of the damage caused by mercury in the brain is due to mercury's ability to cause oxidative damage to not only the proteins and lipids in the brain but also to DNA. Discuss with your clinician the presence of variants along with past and present mercury exposures to help guide treatment.

Heavy Metals Detox & Glutathione Risk SNPs

Gene Name	GCLC
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	9.6% of people have this result
Impact	High

"Glutamate-Cysteine ligase catalytic subunit" gene

This variant leads you to have a decreased ability to make an important antioxidant called glutathione. This can decrease your body's ability to clear toxins, pesticides, etc., which can lead to various health concerns.

- This variant is associated with:
 - Impaired mercury detox
 - Increased risk of psychosis
 - Increased size of stroke damage
 - Allergic-type reaction to sulfamethoxazole

Supplements: Supplements like glutathione, sulforaphane, and N-acetylcysteine (NAC) can help support glutathione synthesis and can be considered with guidance on dosing.

Avoid: Avoid glyphosates, as they can further increase risk, as supported by this SNP's pathway.

Gene Name	GSTP1
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	11.1% of people have this result
Impact	High

"Glutathione S-Transferase Pi 1" gene

Glutathione is an enzyme that is important for the removal of mercury and other toxins, as well as chemicals from the body. This variant in the glutathione-related gene (GSTM1) results in a lower ability to get rid of heavy metals, pesticides, and other toxins such as smoke by-products. This can lead to high mercury and other toxin levels. Lower ability to detox can contribute to a greater risk of cognitive decline, birth defects, and other health conditions when exposed to glyphosates (RoundUp), pesticides, and other toxins.

- Glutathione is the master antioxidant for the body, important for detoxification.
- Minor allele G is associated with elevated mercury levels and excess susceptibility to the detrimental cognitive effects of cumulative lead exposure.
- Minor allele G is associated with difficulties removing glyphosate (RoundUp) from the body. The study showed G allele had a 2.6 times the risk for miscarriages and birth defects in infants born to mothers with VERY high glyphosate exposure. If two variants (GG) 4.8 times the risk.
- Two variants (GG) have been associated with nearly 50% decreased ability to clear smoke by-products from the body and a 1.84 times the risk for cognitive impairment/ Alzheimers.
- GSTP1 plays an important role in detoxifying carcinogens that can lead to cancer risks.
- Involved in over 200 different diseases.

Diet: Getting optimal amounts of Vitamin C, via dietary options, and avoiding glyphosate exposure from non-organic sources. See the "Clean 15 and Dirty Dozen" and "Glyphosate - Foods & clearing toxins" handout.

Supplements: Supplementing with products such as sulforaphane, N-acetylcysteine (NAC), direct glutathione replacement, and vitamin C can help to support this pathway and your ability to remove toxins from the body.

Gene Name	NFE2L2
Minor Allele	T
My Results	TG
My Variants	You have 1 minor allele
Prevalence	21.4% of people have this result
Impact	Medium

"Nuclear Factor Erythroid 2-Related Factor 2" gene

With this variant, you may have less ability to turn on various detox pathways, such as many of the glutathione pathways above. This can lead to increased oxidative stress (i.e. more "rusting" or "tarnishing" of the cells) and higher susceptibility to the damaging effects of different toxic chemicals and substances.

- Codes for the transcription factor Nrf2 (master regulator of detoxification/ inflammation - acting as a switch), which turn on many other genes, which are important for reducing oxidative stress.
- Variants have been associated with decreased Nrf2 expression.
- May be at greater risk for cerebrovascular disease (a disease that affects the blood vessels and blood supply to the brain).
- Two variants (TT) convey a lower risk for hypertension or elevated blood pressure.

Supplements: Sulforaphane, hydrogen water, resveratrol and cordyseps can be beneficial options to support this variant.

Note: Lowering inflammation, in general, can support this pathway as well.

Gene Name	ABCC1
Minor Allele	C
My Results	CT
My Variants	You have 1 minor allele
Prevalence	48.7% of people have this result
Impact	Medium

"ATP Binding Cassette Subfamily C Member 1" gene

This gene is important for the ability to transport mercury, mold-related toxins, and other toxins out of the brain. With this variant, you may have less of an ability to remove these toxins from the brain and can result in a greater risk of mercury toxicity in the brain. Additionally, this variant can increase the risk for neurodevelopmental disorders in children born to mothers consuming high amounts of high mercury foods, and may also influence mold-related cognitive issues.

- Involved in transport and elimination of drugs and toxins such as mercury.
- Carriers of the C allele less able to remove mercury from the brain and nervous tissue.
- Individuals with two variants (CC) has been associated with:
 - Higher mercury **tissue** levels with higher fish intake than individuals with TT variants.
 - Lower neurodevelopmental scores in children born to mothers who consumed a high fish diet. A particular type of mercury (methyl mercury) easily crosses the blood-placenta barrier and accumulated more in the fetus than in the mother.

Diet: You may find benefit from limiting the intake of high mercury fish king mackerel, shark, tilefish, swordfish, bigeye tuna, marlin, orange roughy etc.), particularly if 2 variants (AA).

Supplements: Sulforaphane, NAC (particularly if high mercury exposure from fish) may be beneficial to support this pathway.

Gene Name	GSTM1
Minor Allele	A
My Results	A
My Variants	You have 2 minor alleles
Prevalence	50.9% of people have this result
Impact	High

"Glutathione S-Transferase Mu 1" gene

Glutathione is an enzyme that is important for the removal of mercury and other toxins, as well as chemicals from the body. You are completely missing this glutathione-related gene (GSTM1) due to this deletion. Decreased ability to clear these toxins can contribute to many health conditions including the risk of certain cancers, Alzheimer's, hardening of the arteries (atherosclerosis), and asthma.

- Important enzyme in the glutathione detox system. Glutathione is the master antioxidant for the body, important for detoxification.
- Individuals with the "A" variant have a complete deletion of this gene and it is associated with increased oxidative stress on cells and lower ability to clear toxins (mercury, glyphosate/pesticides) from cells.
- Also associated with risk of Alzheimer's, atherosclerosis, asthma, and lung cancer development as well as low birth weight for babies of women with this SNP.
- The combination of multiple variants in the other glutathione-related genes can have an additive effect for the risk of conditions such as lung cancer.

Diet: Getting optimal amounts of Vitamin C, via dietary options, and avoiding glyphosate exposure from non-organic sources. See the "Clean 15 and Dirty Dozen" and "Glyphosate - Foods Clearing Toxins" handout.

Supplements: Supplementing with products such as sulforaphane, N-acetylcysteine (NAC), direct glutathione replacement, and vitamin C can help to support this pathway and your ability to remove toxins from the body.

Anxiety and Stress Vulnerability

Anxiety can manifest in many different ways. Some general symptoms of anxiety include persistent and excessive worrying, repetition of words or phrases, or fixating on worrying. Symptoms can include episodes of panic. Panic can manifest with physical symptoms such as rapid heart rate, sweating, along with doom-type symptoms or fears. More extreme forms of anxiety manifest as phobias (an extreme or irrational fear). Generalized Anxiety Disorder (GAD), panic disorder, agoraphobia (fear of leaving one's home or crowded places), social phobia, and specific phobias can have some overlapping and some very different contributing factors.

The genomics of anxiety are complex, but this panel looks at small changes in DNA called SNP's or variants that relate to brain chemicals like serotonin ("happy and calming" brain chemical), norepinephrine/adrenaline (involved in the "fight or flight" response), dopamine (the "feel good" brain chemical), cortisol (the "stress" hormone), and more. We want the right balance of these brain chemicals. Variants can result in abnormalities in these brain chemical levels, expression, and activation, resulting in greater risk for anxiety and anxiety-like symptoms. Anxiety disorders are prevalent in the world, and it has been estimated that approximately one-third of people will develop anxiety at some point in their lives. Genomics can provide insight into one's risk factors as well as personalized interventions to assist you.

Anxiety and Stress Vulnerability Risk SNPs

Gene Name	ADRA1A
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	6.5% of people have this result
Impact	High

"Alpha-1-adrenergic Receptor" gene

Individuals with this SNP may have an increased adrenaline (fight or flight hormone) response which can increase the risk for anxiety, especially in older patients.

- In individuals aged 65 and over, 2 variants have been associated with over 4 times the risk for generalized anxiety disorder (GAD).
- You may have an increased risk of ADHD.

Diet: Maintaining a low carbohydrate and low sugar diet may help. Incorporating magnesium into your diet may reduce anxiety levels.

Lifestyle: Meditation or yoga can be beneficial with this variant.

Supplements: Magnesium, vitamin B6, ashwagandha, L-theanine, or magnolia may help decrease your adrenaline response and can be used with guidance for proper dosing.

Gene Name	RGS2
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	7.8% of people have this result
Impact	High

"Regulator of G Protein Signaling 2" gene

Serotonin is the happy, calm brain chemical. You may make less of a protein that is important for serotonin release. This leads to less serotonin release from your neurons and lower serotonin signaling in parts of the brain that regulate anxiety.

- This is associated with a slightly increased risk for panic disorder as well as an increased risk of generalized abdominal pain.
- You may have a 50% less chance of responding to sertraline (antidepressant/SSRI medication) treatment in patients with anxiety.

Diet: Eating a diet rich in tryptophan may be beneficial. Tryptophan-rich foods include turkey, chicken, dairy, fish, egg white, pumpkin seeds, and peanuts.

Supplements: Extended-release 5-HTP, and forskolin have evidence in helping improve this pathway and can be considered with guidance on dosing, risks, and potential benefits.

Medications: Oxytocin may be beneficial. You may be less likely to benefit from SSRIs (fluoxetine, escitalopram, sertraline, etc.) than typical. Your clinician may also want you to avoid the medication Xyrem (commonly prescribed for narcolepsy) as it lowers the activity of this protein even more.

Gene Name	ESR2
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	12.2% of people have this result
Impact	High

"Estrogen Receptor Beta" gene

Stimulation of estrogen receptors can help decrease anxiety levels. You may have less estrogen receptor activity which can lead to anxiety and anxiety-like symptoms.

- In women over 65 years of age, this variant has been associated with double the risk for generalized anxiety disorder (GAD).
- For women not on hormone replacement, this variant may also increase the risk of depression.

Supplements: Genistein, resveratrol (in red grapes), pterostilbene (in blueberries), and rhubarb extract (ERr 731) can help improve estrogen levels but should be used with guidance for proper dosing, risks, and potential benefits.

Medications: Progesterone supplementation can make anxiety worse, so if you are considering hormone replacement, you may want to let your clinician know.

Gene Name	ADRA1A
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	16% of people have this result
Impact	High

"Alpha-1-adrenergic receptor" gene

Individuals with this SNP may have an increased adrenaline (fight or flight hormone) response which can increase the risk for anxiety, especially in elderly patients.

- A study showed that people over the age of 65 have 4.48 times increased risk of generalized anxiety disorder (GAD).
- You may have an increased risk of ADHD.

Diet: Maintaining a low carbohydrate and sugar diet may help.

Incorporating magnesium into your diet may also be beneficial in reducing anxiety levels.

Lifestyle: Meditation or yoga can be beneficial with this variant.

Supplements: Magnesium, vitamin B6, ashwagandha, L-theanine, or magnolia may help decrease your adrenaline response and can be used with guidance for proper dosing.

Gene Name	NPSR1
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	19.6% of people have this result
Impact	High

"Neuropeptide S Receptor 1" gene

Individuals with this SNP tend to have overactive "wakefulness" in the brain due to high NPSR1 receptor activity. This can induce anxiety and panic, increased flushing response, and increased histamine release during times of stress.

- Individuals that have had childhood trauma AND who have 2 variants are more sensitive to anxiety.
- This variant can lead to you over-interpreting reactions to fear, as well as increase your risk of having difficulty sleeping when you are stressed.

Lifestyle: You may want to limit caffeine use as it can increase orexin (a brain chemical that this variant may increase) even more.

Supplements: Pycnogenol, quercetin, bromelain, vitamin C, melatonin, and luteolin may be beneficial at helping decrease histamine release and/or "hyperarousal", but should be used with guidance regarding dosing, risks, and potential benefits.

Medications: Belsomra has evidence to decrease orexin (the brain chemical that this variant may increase) if you have insomnia.

Gene Name	BDNF
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	31.1% of people have this result
Impact	Medium

"Brain-Derived Neurotrophic Factor" gene

Individuals with this SNP have lower levels of BDNF (a protein that acts like fertilizer for the brain), which has been associated with anxiety and Obsessive-Compulsive Disorder (OCD).

- This BDNF protein interacts with serotonin (happy brain chemical) to promote the brain's flexibility/plasticity.
- You may have lower levels of BDNF (fertilizer for the brain) and higher cortisol levels (stress hormone).
- This may increase your risk of social anxiety or feelings of nervousness with you undergo physical stress.

Diet: Consuming butyrate-producing foods can raise BDNF levels. Examples of foods that are rich in butyrate include asparagus, garlic, and onions. There is a Butyrate Rich Foods handout found in the Document Library that your clinician can provide you with more examples. High fiber foods may also help raise BDNF levels.

Lifestyle: Moderate to intense exercise, and Transcranial Magnetic Stimulation have been shown to raise BDNF levels and can be discussed with your clinician.

Supplements: Certain prebiotics, silymarin, curcumin, and lithium can raise BDNF levels and can be considered with guidance on dosing.

Medications: There are certain SSRIs (sertraline) that may help anxiety better than others. Estrogen can raise BDNF levels.

Gene Name	ADRB2
Minor Allele	G
My Results	GC
My Variants	You have 1 minor allele
Prevalence	46.1% of people have this result
Impact	Medium

"Beta 2-Adrenergic Receptor"

This SNP increases response to adrenaline (fight or flight hormone) which can be associated with anxiety, as well as GI symptoms.

- This variant is associated with over double the risk of anxiety.
- This can also increase the risk for gastrointestinal (GI) issues, including indigestion.

Diet: Maintaining a low carbohydrate and sugar diet may help. Incorporating magnesium into your diet may also be beneficial in reducing anxiety levels.

Lifestyle: Meditation or yoga can be beneficial with this variant.

Supplements: Magnesium, vitamin B6, ashwagandha, L-theanine, or magnolia may help decrease your adrenaline response and can be used with guidance for proper dosing.

Gene Name	DIO2
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	46.9% of people have this result
Impact	Medium

"Iodothyronine Deiodinase 2" gene

Individuals with this SNP do not convert T4 (storage form) to T3 (active form) well in the brain (even though blood levels may be normal). If you have two variants this can lead to increased symptoms of anxiety or depression.

- If you have two variants you may have an increased risk of anxiety and depression.
- Hypothyroidism has been associated with greater anxiety levels than those with normal thyroid levels.

You may benefit from having your thyroid levels (TSH and T3) checked.

Supplements: If you are hypothyroid replacing T3 with liothyronine, armour thyroid, Naturethyroid, or WP thyroid may be beneficial. If you are not hypothyroid selenium or kaempferol may be beneficial. These can be used with guidance regarding dosing, risks, and potential benefits.

Gene Name	TPH1
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	47.5% of people have this result
Impact	Medium

"Tryptophan Hydroxylase 1" gene

Serotonin is the happy, calm brain chemical. Serotonin is made from the protein building block, tryptophan. You may have difficulty creating serotonin from tryptophan. This can increase your risk for anxiety.

- This variant may increase your risk of panic disorder.

Lifestyle: You may benefit from bright light/light therapy and exercise as this increases serotonin production.

Supplements: Extended-release 5-HTP, SAME, vitamin D, and prebiotics/probiotics have some evidence for increasing serotonin and/or improving this pathway, but should be used with guidance regarding dosing, risks, and potential benefits.

Medications: SSRIs (fluoxetine, escitalopram, sertraline, etc.) can help this pathway and improve serotonin levels.

Gene Name	ADORA2A
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	48.1% of people have this result
Impact	Medium

"Adenosine A2A Receptor" gene

This gene is normally responsible for keeping glutamate (excitatory brain chemical) and dopamine (associated with movement and reward loop) from getting too high. Individuals with this SNP appear to have lower levels of this receptor, leading to higher glutamate and to a lesser extent, higher dopamine.

- This increases your risk of panic disorder and agoraphobia (extreme fear and avoiding places or situations that might cause panic.)
- If you have two variants you are more likely to feel anxious after drinking something with caffeine or after taking a medication with amphetamine in it (ADHD medication).
- This SNP is highly linked with another gene (rs2298383) which has been associated with 3 times the risk of Multiple Chemical Sensitivity (MCS).

Lifestyle: You may want to limit or avoid caffeine with this variant.

Supplements: NAC, L-theanine, and DHA (an omega 3) are examples of supplements that you may benefit from and can be discussed for proper dosing with a clinician.

Gene Name	ADORA2A
Minor Allele	C
My Results	CT
My Variants	You have 1 minor allele
Prevalence	49.9% of people have this result
Impact	Medium

"Adenosine A2A Receptor" gene

This gene helps balance brain chemicals like glutamate and dopamine. With this SNP, glutamate levels can be higher which is associated with increased anxiety, caffeine sensitivity, and sleep disturbances.

- Variants associated with increased risk of seizures.
- If you have two variants you may be more likely to have:
 - Increased anxiety, worsened when exposed to caffeine.
 - Note: Other genetic and clinical factors may also influence a patient's response to caffeine.
 - Increased risk of seizures, comorbid neurological disorders for those with seizures
 - Sleep disturbances and difficulty concentrating

Lifestyle: You may want to limit or avoid caffeine with this variant.

Supplements: NAC, L-theanine, Genistein, and DHA (an omega 3) are examples of supplements that you may benefit from and can be discussed for proper dosing with a clinician.

Gene Name	HTR1A
Minor Allele	G
My Results	CG
My Variants	You have 1 minor allele
Prevalence	49.9% of people have this result
Impact	Medium

"5-Hydroxytryptamine Receptor 1A" gene

Serotonin is a very important chemical in the brain which helps you feel happy and calm. This variant results in lower serotonin, which can be associated with an increased tendency towards both anxiety and depression.

- This can increase your risk of anxiety and depression by 3 times compared to someone who does not have this variant.
- 2 variants may increase the risk of impulsive behavior.

Diet: Eating a diet rich in tryptophan may be beneficial. Tryptophan-rich foods include turkey, chicken, dairy, fish, egg white, pumpkin seeds, and peanuts.

Lifestyle: You may benefit from bright light/light therapy and exercise.

Gene Name	COMT V158M
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	50% of people have this result
Impact	Medium

"Catechol-O-Methyltransferase" gene

Individuals with this SNP are associated with the reduced enzymatic ability to break down brain chemicals such as dopamine ("feel good" chemical) and epinephrine/adrenaline (fight or flight). Increased levels of these can result in symptoms of anxiety.

- This slows your ability to break down dopamine and adrenaline.
- You may have an increased risk of panic disorder.
- If you have two variants you are less likely to respond to SSRIs.
- Women with 2 variants may be more likely to experience episodic anxiety and be more introverted. This was not found in men who have two copies of this variant.

Diet: Limiting sugar is beneficial with this pathway. You may also benefit from foods that contain magnesium, and zinc as they help stimulate this enzyme.

Supplements: Ashwagandha, zinc, magnesium, and curcumin are supplements that may be beneficial for this pathway but should be used with guidance regarding dosing, risks, and potential benefits.

Medications: If you have two copies, you may be less likely to respond to SSRIs for the treatment of anxiety and/or depression.

Note: If you have two copies you may want to avoid quercetin as it can reduce the activity of this enzyme further.

Anxiety and Stress Vulnerability Benefit SNPs

Gene Name	MAOA (R297R)
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	11.7% of people have this result
Impact	Helpful

"Monoamine Oxidase A" gene

Individuals with two variants are associated with a greater ability to break down catecholamines (brain chemicals) such as dopamine (happy, calming brain chemical) and epinephrine/adrenaline (fight or flight hormone). Lower levels of these catecholamines can result in the alleviation of symptoms of anxiety.

- Two variants decrease your risk of generalized anxiety disorder.
- While a benefit for anxiety, this variant is associated with an increased risk of ADHD.

Recommendations if 1 (or 0) variant:

Supplements: Ashwagandha, iron, magnesium, rhodiola, and zinc can help address this pathway and can be used with guidance for proper dosing.

Gene Name	TPH2
Minor Allele	T
My Results	GT
My Variants	You have 1 minor allele
Prevalence	35% of people have this result
Impact	 Helpful

"Tryptophan Hydroxylase 2" gene
Individuals with this SNP have the benefit of making more serotonin (happy and calm brain chemical), resulting in a lower risk for anxiety.

- This variant decreases your risk of panic disorder.
- If you have two variants you have a decreased risk of anxiety.

Gene Name	CRHR1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	49.3% of people have this result
Impact	 Helpful

"Corticotropin Releasing Hormone Receptor 1" gene

If you have two variants you may have lower cortisol (stress hormone) levels which decrease your risk of anxiety.

- Two variants have been associated with lower cortisol (stress hormone) levels compared to those without this variant.
- Two variants also have been associated with decreased risk of GI symptoms related to anxiety.

Recommendations if 1 (or 0) variant:

Lifestyle: Yoga may be beneficial.
Supplements: Resveratrol, iron, taurine, and pterostilbene (from blueberries) are other ways to help address this pathway and can be used with guidance for proper dosing.

Depression

Depression is a widespread disorder that affects millions worldwide. The general symptoms of depression can include persistent sadness and loss of interest or motivation. Depression can present itself in a multitude of ways including angry outbursts, irritability, sleep disturbance, lethargy, loss of appetite, feeling worthless or guilty, and suicidal thoughts or actions. Additionally, depression has been shown to commonly be diagnosed side-by-side with anxiety or other mood disorders. The genomics of depression are complex, and environmental factors must be taken into consideration when talking about the underlying factors of symptoms. This panel looks at small changes in DNA called SNP's or variants. Variants that relate to brain chemicals like serotonin ("happy and calming" brain chemical), dopamine (the "feel good" brain chemical), cortisol (the "stress" hormone), and more are included in this report. The correct balance of these brain chemicals is important. Alterations caused by the genetic variants can change levels of these chemicals lead to structural changes of the brain, and ultimately lead to potential depressive symptoms. Genomics helps provide insight into the risk factors as well as personalized interventions to assist you.

Depression Risk SNPs

Gene Name	GCH1
Minor Allele	A
My Results	AA
My Variants	You have 2 minor alleles
Prevalence	4.1% of people have this result
Impact	High

"GTP cyclohydrolase 1" gene

This gene (GCH) regulates the production of a molecule called BH4. BH4 is important to make serotonin, dopamine, and glutamate. This variant is associated with lower levels of GCH.

- You may have lower serotonin synthesis, which is associated with increased amounts of depressive symptoms.

Diet: A diet rich in nitric oxide (leafy greens, beets, etc.) may be beneficial if you are also experiencing attention and focus difficulties. Ask your clinician for our Nitric Oxide producing foods handout in our document library.

Supplements: Resveratrol, omega 3's, folinic acid, methylfolate, and ginkgo biloba may be beneficial.

Gene Name	BDNF
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	31.1% of people have this result
Impact	Medium

"Brain-Derived Neurotropic Factor" gene

Brain-derived neurotrophic factor (BDNF) is a growth factor, kind of like fertilizer for the brain. This SNP contributes to lower levels of BDNF.

- This variant is associated with increased risk of depression.

Lifestyle: Moderate to intense exercise may improve BDNF levels.

Supplements: Silymarin and saffron have evidence to improve BDNF levels.

Gene Name	HTR2A
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	40.9% of people have this result
Impact	Medium

"5-Hydroxytryptamine Receptor 2A" gene

Serotonin is a "calming or happy" brain chemical. This gene codes for a serotonin receptor, but also influences other neurotransmitters (brain chemicals) in your body that relate to your mood. You may have increased activity of this receptor leading to lower serotonin, lower GABA ("calming" brain chemical), and higher cortisol ("stress" hormone).

- This variant may increase depressive symptoms.
- You are more likely to respond to SSRI and SNRI antidepressants, especially the medication citalopram.

Supplements: L-theanine may help raise GABA levels; ashwagandha, and magnolia extract may help lower cortisol levels; saffron may improve serotonin levels in your brain.

Medications: Your clinician may recommend SSRI's and SNRI's to help improve serotonin signaling.

Gene Name	HTR1A
Minor Allele	G
My Results	CG
My Variants	You have 1 minor allele
Prevalence	49.9% of people have this result
Impact	Medium

"5-Hydroxytryptamine Receptor 1A" gene

Serotonin is a "calming or happy" brain chemical. You are more likely to halt and prevent the release of serotonin. This results in lower serotonin in various regions of your brain.

- You are likely to have decreased serotonin levels.
- This variant increases your risk of depression by almost 3 times.
- If you have two copies of this SNP, you may be prone to more impulsive behavior.

Lifestyle: Bright light and exercise may be beneficial.

Supplements: 5-HTP (extended-release shows greater efficacy), saffron, and SAME are supplements that you may benefit from.

Medications: Your clinician may consider SSRIs and SNRIs, as they are examples of medications that improve serotonin signaling in your body.

Depression Benefit SNPs

Gene Name	ITIH4
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	7.6% of people have this result
Impact	Helpful

"Inter-alpha-trypsin inhibitor heavy chain 4" gene

Complement is an inflammatory pathway that causes cellular destruction and is activated by bacteria, molds and yeasts, viruses, and parasites. 2 variants in this pathway decreases activation of the complement system.

- If you have two copies of this variant you have a decreased risk of depression.

Gene Name	KMO
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	28.9% of people have this result
Impact	Helpful

You have a genomic variant(s) in this "Kynurenine 3-Monooxygenase" pathway.

Key Point: Individuals with this SNP have relative protection from depression by not increasing the number of toxic molecules made in this pathway. The T allele mediates the normal function of this pathway.

- This gene is involved in the tryptophan metabolism, which is important for serotonin production.
- T allele mediates normal serotonin synthesis via this pathway.
- G allele increases the risk for depression by decreasing serotonin levels.

Gene Name	IL6R
Minor Allele	C
My Results	AC
My Variants	You have 1 minor allele
Prevalence	44.5% of people have this result
Impact	Helpful

You have a genomic variant(s) in this "Interleukin 6 Receptor" pathway.
Key Point: Inflammation caused by IL6 has been associated with depression. Individuals with this SNP have an increased ability to decrease brain inflammation caused by IL6, and therefore, have a lower risk for severe depression.

- This gene is involved in the signaling of IL6, which can cause brain inflammation.
- C allele lowers the signaling of IL6, which can lower this brain inflammation, and decrease the risk for depression.

Gene Name	CRHR1
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	49.3% of people have this result
Impact	Helpful

"Corticotropin Releasing Hormone Receptor 1" gene
Cortisol is a hormone that is released in response to stress. Two variants are associated with decreased cortisol in response to stress.

- If you have two variants you have a decreased risk of depression.

Mast Cell and IgE

Mast cells are highly specialized cells that release various chemicals like histamine and other inflammatory signals into the blood vessels as well as the brain and skin. Mast cells are best known for their involvement in allergic disease but also have an essential role as first-line defenders against invading foreign invaders such as parasites, bacteria and other infections.

When activated by a special type of immunoglobulin called IgE, mast cells release histamine, interleukin-13 (IL13), interleukin-4 (IL4), and other molecules. These molecules trigger a cascade of inflammatory symptoms. Due to the release of histamine when mast cells are activated, symptoms of mast cell activation can include flushing, itching or hives, depression, mood changes, problems concentrating, abdominal pain, diarrhea, nausea, and/or vomiting. Mast cell activation syndrome (MCAS) describes the illness and symptoms triggered when mast cells release too many of these mediating substances that cause allergy-like symptoms. Systemic Mastocytosis (SM) is different than MCAS. It is a disease that is most often due to a rare SNP in the KIT gene causing extreme overactivation of mast cells. Individuals with a KIT variant (rare) should discuss this with a specialist as KIT can be life-threatening if left unchecked. With systemic mastocytosis, gastrointestinal symptoms may be a presenting sign 60-80% of the time. Other rare symptoms associated with mast cell disease include anemia, bone or muscle pain, and enlarged liver, spleen and lymph nodes.

It is important to note when addressing mast cell and histamine-related pathways that histamine is not always bad. It serves important functions in the brain as an excitatory (wakefulness) neurotransmitter. However overactive release of histamine can be problematic.

There are 2 main receptors on the surface of mast cells, FCER1 (high-affinity) and FCER2 (low-affinity). These can contribute to allergic reactions and even to a paradoxical response to steroids. There are lots of natural interventions that can help stabilize mast cells and lower histamine release. Antihistamines, in spite of their name, do not lower histamine release. Antihistamines, instead they block the binding of histamine to certain receptors but do not decrease the release of interleukins (inflammation) triggered by allergic pathways and so sometimes do not work fully to control symptoms.

Mast Cell and IgE Risk SNPs

Gene Name	IL4
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	4.9% of people have this result
Impact	High

"Interleukin 4" gene

Individuals with this SNP may be prone to increased IL4, allergy cell activation, and allergic responses. IL4 is an inflammatory molecule produced by immune and allergy cells ("mast cells") that triggers more production of itself, IL13, and ultimately IgE. This stimulates and maintains allergic inflammatory responses.

- You may have an increased risk of asthma and allergic conditions such as eosinophilic esophagitis.

Supplements: NAC (N-acetylcysteine), quercetin, apigenin, vitamin E, green tea extract, and luteolin can help lower IL4, mast cell degranulation, and/or IgE responses.

Gene Name	TSLP
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	14.4% of people have this result
Impact	High

"Thyroid Stromal Lymphopoietin" gene

TSLP is part of an inflammatory pathway expressed in the body's protective linings of the nose, esophagus, lungs, gut, and blood vessels. Individuals with this SNP produce more of this protein which can contribute to asthma, as well as hives and even heart disease

- You may have an increased risk of asthma and hives

Supplements: Resveratrol, curcumin, omega 3's, and lipoic acid may benefit this pathway.

Medications: Your clinician may discuss other prescription options if you have severe symptoms.

Gene Name	FCER2
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	24.1% of people have this result
Impact	Medium

"Fc Epsilon Receptor II" gene

FCER2 is the weaker receptor for IgE that is commonly found on the smooth muscles of airways and on B cells. FCER2 plays a role in regulating IgE and its signaling. This variant causes the receptor to bind IgE more strongly, creating downstream effects that lead to increased responsiveness to IL4 and higher IgE production.

- This variant conveys an increased risk of asthma and severe exacerbations in asthmatics.

Supplements: Piceid resveratrol, kaempferol may be beneficial as they can help lower downstream effects of high IgE. Vitamin E and green tea extract can help lower IgE. Supplements that block mast cell degranulation such as perilla extract, pycnogenol, ellagic acid, and N-acetylcysteine may be considered with guidance on dosing.

Medications: Your clinician may discuss Ketotifen which is an asthma medication that stabilizes mast cells.

Gene Name	HRH1 T-17C
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	28.2% of people have this result
Impact	Medium

"Histamine Receptor H1" gene

HRH1 is one of the receptors for histamine. Histamine mediates allergic responses but also contributes to the sleep-wake cycle, cognitive function, anxiety, and more. This variant is associated with lower HRH1 levels, which has been shown to enhance certain pathways that make allergic responses worse.

- You may have an increased risk of allergic rhinitis and a higher likelihood of developing asthma.
- Antihistamines such as claritin and zyrtec tend to work less well in individuals with 2 copies of this variant.

Medications: Salmeterol, formoterol, arformoterol may benefit this pathway. Typical antihistamines (claritin, zyrtec, etc.) might be less effective with 1-2 copies of this SNP - consult with your clinician.

Gene Name	FCER2
Minor Allele	G
My Results	AG
My Variants	You have 1 minor allele
Prevalence	33% of people have this result
Impact	Medium

"Fc Epsilon Receptor II" gene

FCER2 is the weaker receptor for IgE that is commonly found on the smooth muscles of airways and on B cells. FCER2 plays a role in regulating IgE and its signaling. This variant decreases FCER2 expression which ultimately messes up the checks and balances system regulating IgE production, resulting in higher IgE levels.

- This variant conveys an increased risk of COPD and uncontrolled asthma due to decreased steroid responsiveness. Steroids may make things worse as the highest IgE levels were found in GG (2 variants) individuals taking steroids.
- This SNP is associated with a higher risk of severe exacerbations in asthmatic children.

Medications: Your clinician may exercise caution with steroid use (as can make asthma worse, particularly if GG) and instead might recommend montelukast for asthma control.

Supplements: Piceid resveratrol and kaempferol may be beneficial as they can help lower downstream effects of high IgE. Vitamin E and green tea extract can help lower IgE and may be considered with guidance on dosing.

Gene Name	CXCL8
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	35.7% of people have this result
Impact	Medium

"C-X-C Motif Chemokine Ligand 8" gene

CXCL8/IL8 plays a role in the body's response to infection, tissue damage and inflammation as it guides immune cells towards these areas of injury. Individuals with this SNP may be prone to increased IL8 levels and more tissue damage. Higher levels of IL8 are seen in individuals with moderate to severe asthma.

- TT (2 variants) conveys higher levels of IgE.

Supplements: Quercetin, fistein, and cats claw all lower IL8 and may be considered with guidance on dosing.

Lifestyle: Exercise and disruptions in prolonged sitting may benefit this pathway.

Medications: Your clinician may discuss azelastine as it lowers IL8.

Gene Name	IL13
Minor Allele	T
My Results	CT
My Variants	You have 1 minor allele
Prevalence	38.9% of people have this result
Impact	Medium

"Interleukin 13" gene

Individuals with this SNP are prone to higher levels of IL13 and IgE. IL13 is a key player in the inflammatory response triggered by an invading parasite or allergen in the lungs and skin. IL13 triggers immune cells to make more of the immunoglobulin IgE, which drives allergic responses.

- You may have a higher risk of allergies, asthma and COPD.

Supplements: Rosemary extract, quercetin, omega 3's, luteolin, vitamin E, and green tea extract can help lower IL13 and/or IgE allergic responses.

Medications: Dexamethasone and other steroids, Montelukast may be discussed with your clinician.

Gene Name	IL13
Minor Allele	A
My Results	AG
My Variants	You have 1 minor allele
Prevalence	40.3% of people have this result
Impact	Medium

"Interleukin 13" gene

Individuals with this SNP are prone to higher levels of IL13 and IgE. IL13 is a key player in the inflammatory response triggered by an invading parasite or allergen in the lungs and skin. IL13 triggers immune cells to make more of the immunoglobulin IgE, which drives allergic responses.

- You may have a higher risk of allergies, asthma, COPD, and less of a response to asthma treatment (budesonide inhalation suspension combined with fluticasone propionate).

Supplements: Rosemary extract, quercetin, omega 3's, luteolin, vitamin E, and green tea extract can help lower IL13 and/or IgE allergic responses.

Medications: Dexamethasone, Montelukast, or Monoclonal antibodies targeting IL13 or IgE may be discussed with your clinician.

Gene Name	FCER1A
Minor Allele	T
My Results	TC
My Variants	You have 1 minor allele
Prevalence	47.7% of people have this result
Impact	Medium

"Fc Fragment of IgE Receptor 1A" gene

Individuals with this SNP may be more prone to allergies and hypersensitivity reactions due to higher levels of FCER1A and IgE. FCER1A is the IgE receptor found on mast cells, and its primary function is to kick off immune reactions in response foreign invaders or allergens. By increasing both FCER1A and IgE, this variant contributes to more mast cell activation (degranulation) and worse allergic reactions.

- You may have an increased risk of asthma, eczema, and exaggerated responses to environmental allergens like dust mites.

Supplements: Piceid resveratrol, kaempferol may be beneficial as they can help lower FCER1A. Your clinician may also discuss other supplements that block mast cell degranulation such as isoquercetin, NAC, ellagic acid, perilla extract, and apigenin.

Gene Name	CXCL8
Minor Allele	T
My Results	AT
My Variants	You have 1 minor allele
Prevalence	49.6% of people have this result
Impact	Medium

"C-X-C Motif Chemokine Ligand 8" gene

CXCL8/IL8 plays a role in the body's response to infection, tissue damage and inflammation as it guides immune cells towards these areas of injury. Individuals with this SNP are prone to increased IL8 levels and more tissue damage. Higher levels of IL8 are seen in individuals with moderate to severe asthma.

- You may have an increased risk of developing asthma.

Supplements: Quercetin, fistein, and cats claw all lower IL8 and may be considered with guidance on dosing.

Lifestyle: Exercise and disruptions in prolonged sitting may benefit this pathway.

Medications: Your clinician may discuss azelastine as it lowers IL8.

Obesity and Weight Control

Obesity is the result of complex interactions between genetics and environmental factors such as diet, sleep, mental and physical health and level of physical activity. Current evidence suggests that about 40-70% of susceptibility to obesity is explained by genetics. Additionally, specific diets combined with genetics can alter an individual's metabolism, independent of caloric intake. Targeting dietary style to genomics can in some cases help patients be more successful at long term weight loss and less susceptible to regaining weight.

The best understood gene risk variants, effect the ability of your brain to recognize that you have had enough to eat. This lack of feeling satisfied translates to increased caloric intake. Other variants relate to what foods you tolerate well and how you respond to exercise. There is a cumulative effect when multiple variants are present.

You will notice that many of the pathways have similar interventions, because they all interact and ultimately influence the appetite center in the brain and contribute to something called "leptin resistance". Leptin is the bodies normal signal to feel full. Leptin resistance is triggered by saturated fat. Saturated fat is the fat from beef, cheese, pork, and other animal fats as well as from coconut oil. Leptin resistance is not triggered by unsaturated fat (fish, nuts, seeds, olive oil). Healthy sources of fiber are a benefit in preventing leptin resistance (i.e. vegetables).

Additionally, several genes have been shown to have a direct interaction with artificial sweeteners as well as sugar, even when corrected for caloric intake. There are also some general prevention measures that have been shown to be beneficial that are listed in our panel. Some may be surprising like increased coffee intake (decaf or regular). Your clinician will review these with you and may also talk about supplements and or medications that may be helpful with weight loss management. Since there are so many emotional and other factors involved in weight control, working closely with a dietitian may be beneficial.

NOTE: Environmental triggers such as exposure to BPA found in plastic bottles, food containers (microwave meals and water bottles) and the lining of cans, have been shown to affect obesity genes and turn on hunger pathways.

Obesity and Weight Control Risk SNPs

Gene Name	GNB3
Minor Allele	T
My Results	TT
My Variants	You have 2 minor alleles
Prevalence	11.7% of people have this result
Impact (Odds Ratio)	

"G Protein Subunit Beta 3" gene

Individuals with this SNP have reduced breakdown of fat cells. This can lead to the body retaining fat, and increase risk of obesity, especially in men.

- Associated with increased risk of obesity, with stronger effects noted in males (increased risk for both 1 and 2 variants).
- Odds ratio for general population was 1.22 for 2 variants (1 variant was not significant).
- 2 variants associated with increased leptin levels.

Diet: A low saturated fat diet may be beneficial.

Gene Name	COL6A5
Minor Allele	A
My Results	GA
My Variants	You have 1 minor allele
Prevalence	13.9% of people have this result
Impact (Odds Ratio)	High 2.90

"Alpha-5 Containing Collagen VI" gene

Individuals with this SNP have an altered type of collagen, that is more expandable and allows for more fat storage. This can lead to a higher risk of a high BMI and obesity, especially in children.

- This gene is involved in fat tissue's expandability.
- Associated with increased risk for higher BMI.

Lifestyle: This is a tough SNP for intervening once you have gained weight, but theoretically you may respond well to some of the cryotherapy options that shrink the fat cells (adipocytes).

Gene Name	STAT3
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	15.7% of people have this result
Impact (Odds Ratio)	Medium 1.95

"Signal Transducer and Activator of Transcription 3" gene

Individuals with this SNP gain abdominal weight at a markedly higher rate in response to saturated fat. This SNP interacts with the hunger center of the brain and tends to trigger overeating due to resistance to leptin (turns off the feeling of hunger).

- This variant leads to increased weight gain, abdominal obesity and increased insulin resistance.
- For this particular STAT3 variant it is not quite as strong of a genomic risk factor so mostly conveys risk when two copies are present.
- A diet very low in saturated fat (but not necessarily total fat) has been shown to overcome this variant. Keeping saturated fat (found in red meat, cheese, pork and other animal fat) to less than 8% of total calories consumed will prevent the weight gain and obesity associated with this gene.

Diet: A diet low in saturated fat is important with this variant. Ideally, you want to keep your saturated fat to less than 10% of your total calories. A diet rich in omega 3's is a better source of fat (nuts, seeds, avocados, etc.).

Supplements: Fisetin, gynostemma, hesperidin, berberine, and curcumin are supplements that can help with weight loss and can be considered with guidance on dosing.

Medications: Metformin is a prescription medication often prescribed for diabetes that can also help support weight loss.

Gene Name	LEPR
Minor Allele	G
My Results	GG
My Variants	You have 2 minor alleles
Prevalence	21.1% of people have this result
Impact (Odds Ratio)	Medium 1.65

"Leptin Receptor" gene

Leptin is a molecule responsible for inhibiting the feeling of hunger after eating. Individuals with this SNP are less sensitive to leptin, resulting in less ability "turn off" the feeling of hunger in the brain, often resulting in overeating and increased risk of obesity. This SNP has also been shown to interact with a high saturated fat diet to further increase obesity risk.

- This variant can result in leptin insensitivity. Leptin is the signal your body releases to tell you that you are full. Thus, with variants in this pathway, you may feel hungry even after eating.

Diet: A diet low in saturated fat is beneficial to help decrease your risk of obesity.
Medications: There is evidence that oxytocin can help decrease leptin resistance which can help make you feel full after eating.

Gene Name	LEP
Minor Allele	C
My Results	CC
My Variants	You have 2 minor alleles
Prevalence	29.8% of people have this result
Impact (Odds Ratio)	Medium 1.83

"Leptin" gene

Leptin is a hormone made in response to eating and released from stored fat that is meant to curb the feeling of hunger. This SNP contributes to chronically increased leptin, and can result in leptin resistance and increased risk of obesity.

- Leptin normally functions in energy regulation by reducing hunger. This variant is associated with less ability to signal feeling full.
- Suspected of being involved in increased BMI across ethnic groups.

Diet: A diet low in saturated fat can help decrease your risk for obesity. There is evidence that an initial low saturated fat but ketogenic diet followed by a Mediterranean diet can also help reduce your obesity risk.
Supplements: EGCG may help with weight reduction and can be considered with guidance on dosing.

Gene Name	MC4R
Minor Allele	C
My Results	TC
My Variants	You have 1 minor allele
Prevalence	35.9% of people have this result
Impact (Odds Ratio)	Medium 1.30

"Melanocortin 4 Receptor" gene

MCR4 is a receptor in the brain that binds our natural signal that we are full (leptin). Individuals with this SNP have less of these docking sites for leptin and cannot tell when they are full. This leads to excessive eating and snacking.

- By having fewer of these receptors in your brain's hypothalamus you are less able to feel full.
- This gene is associated with weight gain starting in childhood due to increased hunger and more snacking.

Diet: A diet with decreased saturated fat, and rich in omega 3's can help reduce the risk of obesity. You can also consider working with a dietician who can help give guidance to you on your nutritional requirements.

Lifestyle: Exercise helps reduce your risk of obesity. If you have difficulty with overeating you could also consider talking to a counselor to help address these behaviors.

Supplements: EGCG may help with weight reduction and can be considered with guidance on dosing.

Gene Name	STAT3
Minor Allele	G
My Results	GC
My Variants	You have 1 minor allele
Prevalence	39.3% of people have this result
Impact (Odds Ratio)	High 2.22

"Signal Transducer and Activator of Transcription 3" gene

Individuals with this SNP gain abdominal weight at a markedly higher rate in response to saturated fat. This SNP interacts with the hunger center of the brain and tends to trigger overeating due to resistance to leptin (turns off the feeling of hunger).

- This variant leads to increased weight gain, abdominal obesity and increased insulin resistance.
- A diet very low in saturated fat (but not necessarily total fat) has been shown to overcome this variant. Keeping saturated fat (found in red meat, cheese, pork and other animal fat) to less than 8% of total calories consumed will prevent the weight gain and obesity associated with this gene.
- Having more than 2 STAT3 variants or STAT5A conveys the most risk for obesity.

Diet: A diet low in saturated fat is important with this variant. Ideally, you want to keep your saturated fat to less than 10% of your total calories. A diet rich in omega 3's is a better source of fat (nuts, seeds, avocados, etc.).

Supplements: Fisetin, gynostemma, hesperidin, berberine, and curcumin are supplements that can help with weight loss and can be considered with guidance on dosing.

Medications: Metformin is a prescription medication often prescribed for diabetes that can also help support weight loss.

Gene Name	STAT3
Minor Allele	G
My Results	AG
My Variants	You have 1 minor allele
Prevalence	48.8% of people have this result
Impact (Odds Ratio)	High 2.08

"Signal Transducer and Activator of Transcription 3" gene

Individuals with this SNP gain abdominal weight at a markedly higher rate in response to saturated fat. This SNP interacts with the hunger center of the brain and tends to trigger overeating due to resistance to leptin (turns off the feeling of hunger).

- This variant leads to increased weight gain, abdominal obesity and increased insulin resistance.
- A diet very low in saturated fat (but not necessarily total fat) has been shown to overcome this variant. Keeping saturated fat (found in red meat, cheese, pork and other animal fat) to less than 8% of total calories consumed will prevent the weight gain and obesity associated with this gene.
- Having more than 2 STAT3 variants or STAT5A conveys the most risk for obesity.

Diet: A diet low in saturated fat is important with this variant. Ideally, you want to keep your saturated fat to less than 10% of your total calories. A diet rich in omega 3's is a better source of fat (nuts, seeds, avocados, etc.).

Supplements: Fisetin, gynostemma, hesperidin, berberine, and curcumin are supplements that can help with weight loss and can be considered with guidance on dosing.

Medications: Metformin is a prescription medication often prescribed for diabetes that can also help support weight loss.